CONSENTLEDGER: A CONSENT LIFECYCLE AND DATA PRINCIPAL RIGHTS MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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(240701487)

in partial fulfilment of the End Semester Examination for the course

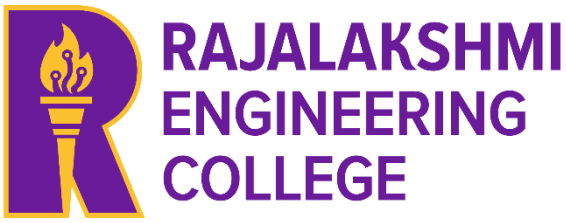
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LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



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BONAFIDE CERTIFICATE

Certified that this project report titled "CONSENTLEDGER: A CONSENT LIFECYCLE AND DATA PRINCIPAL RIGHTS MANAGEMENT SYSTEM" is the bonafide work of SHANTHOSH S (240701487), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, SHANTHOSH S. (240701487), hereby declare that the project report titled "CONSENTLEDGER: A CONSENT LIFECYCLE AND DATA PRINCIPAL RIGHTS MANAGEMENT SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and prediction model — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

Date: 11 November 2026

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SHANTHOSH S

ABSTRACT

Organisations that process personal data must be able to show whose data they hold, what consent was obtained, how requests from individuals were handled and how security incidents were escalated. In many institutions, these records are scattered across spreadsheets, e-mail threads and paper files, making consent status difficult to verify, rights requests easy to miss and breach incidents slow to escalate. This project presents ConsentLedger, a consent lifecycle and data principal rights management system developed using the Zoho Creator low-code platform as an academic prototype in the context of India's Digital Personal Data Protection (DPDP) Act.

The application consists of six forms: Person_Register, Consent_Records, Rights_Requests, Breach_Incidents, Action_Log and Rule_Configuration, supported by reports and a dashboard containing key performance indicator cards and charts. Deluge scripting is used to classify rights requests, assess deadline risks, calculate numerical risk scores for breach incidents and record selected actions in the action log. A prediction model named ConsentLedger_Breach_Risk, trained using a synthetic dataset of 100 records, was deployed on the Breach_Incidents form to generate a predicted risk category for submitted incidents. The ML_High_Risk_Email_Alert workflow sends an e-mail notification when the predicted category is High. The prediction model and notification workflow were tested in the development environment, and an initial sender-address validation issue was resolved by using an address recognised by Zoho Creator.

Since the model was trained using illustrative, rule-generated labels, the accuracy displayed during training does not establish real-world predictive performance. Independent validation using representative data is necessary before any operational use. The project does not claim legal certification, production deployment or an immutable audit trail. Instead, it demonstrates how consent management, data principal rights handling, breach assessment and conditional escalation can be implemented within a single low-code application with minimal hand-written code.

PROJECT AT A GLANCE

Attribute	Details
Project Title	ConsentLedger: A Consent Lifecycle and Data Principal Rights Management System
Application Domain	Privacy Operations and Consent Lifecycle Management (Data Protection – DPDP Act Context)
Student Name & Register No.	Shanthosh S. – 240701487
Zoho Creator Edition	Education
Application Name	ConsentLedger
Application / Portal URL	https://shanthoshs2024csera765.zohocreatorportal.in(portal access and public data-principal login have not been verified).
Number of Forms / Subforms	6 Forms / 0 Subforms
Number of Reports (by type)	7 reports: Person_Register Report, Consent_Records Report, Consent Records List, Rights Requests List, Breach_Incidents Report, Action_Log Report and Rule Configuration Report.
Number of Pages / Dashboards	1 Dashboard (ConsentLedger_Dashboard) containing KPI cards and charts.
Number of Workflows	7 workflow automations: rights-request classification; rights-request deadline assessment on creation and reassessment on Due_Date and Request_Status edits; breach risk-score calculation; consent-creation logging; and conditional high-risk e-mail notification.
Deluge Functions / Custom Actions	Deluge workflow scripts implement automated rights-request classification, deadline-risk assessment, numerical breach risk-score calculation and consent creation logging.
Integrations	No external integrations. Built-in Zoho Creator prediction and email notification capabilities are used.
Roles & Profiles Defined	Application administrator and authorised Zoho Creator users. Separate custom role profiles are not configured.
Environments Used	Development environment. The Zoho Creator Environments feature is not enabled.
Date of Production Deployment	Not applicable. The application has not been deployed to a separate Production environment.

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LIST OF ABBREVIATIONS

The abbreviations used in this report are listed in alphabetical order.

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CO	Course Outcome
CRUD	Create, Read, Update, Delete
DPDP	Digital Personal Data Protection
ER	Entity–Relationship
FR	Functional Requirement
ID	Identifier
KPI	Key Performance Indicator
ML	Machine Learning
NFR	Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
TC	Test Case
UAT	User Acceptance Testing
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1

INTRODUCTION

1.1 Background

Almost every service that an organisation offers today depends on personal data. Student records, customer details, contact information and transaction histories are collected, stored and shared through digital systems, and each of these records belongs to an identifiable individual. India's Digital Personal Data Protection (DPDP) Act, 2023 [1] sets out a framework in which an individual, called the Data Principal, gives consent for the processing of personal data, and an organisation that decides why and how the data is processed, called the Data Fiduciary, carries defined obligations. Those obligations include maintaining the consent that was given, responding to requests that Data Principals make about their data, and taking action when a personal data breach occurs.

For a small organisation, such as a college department or a small enterprise, the operational side of this work is usually informal. Consent is collected on paper or through ad hoc forms, requests from individuals arrive by e-mail or phone and are tracked in a spreadsheet, and a security incident is reported verbally to whoever is available. Nobody has a single view of which individuals have given consent, which requests are close to their deadline, or which incidents deserve urgent attention. ConsentLedger was conceived as a prototype that brings these three activities, consent records, rights requests and breach incidents, into one application built on Zoho Creator.

ConsentLedger is an academic prototype. It demonstrates how selected privacy-management workflows can be built on a low-code platform. It is not a certified compliance product, and the legal obligations of any real organisation would still have to be validated by qualified professionals before the application is used for that purpose.

1.2 Problem Statement

Information about individuals, the consent they have given, the requests they have raised and the security incidents that affect their data is not managed in one place. Because the records are scattered, an organisation cannot easily answer basic questions: whose consent is on record, which requests are overdue or close to their deadline, and which breach incidents are most urgent. Decisions about the severity of an incident are

taken by individual judgement, and urgent incidents depend on a person remembering to inform the right officer. The specific pain points are listed below.

- Personal data principals and their consent records are kept in separate spreadsheets or paper files, so the current consent position is difficult to verify.
- Rights requests are received through different channels, are classified manually and have no consistent record of how close each request is to its deadline.
- Breach incidents are described in free text, and their severity is judged subjectively without a repeatable scoring method.
- There is no automatic escalation, so a serious incident may reach the responsible officer late.
- Actions taken in the system are not recorded in a common log, and management has no dashboard summarising consent, requests and incidents.

1.3 Objectives

The objectives of the project are as follows.

- To design a data model and forms for data principal registration, consent records, rights requests, breach incidents, action logging and rule configuration.
- To automate rights request classification, deadline-risk assessment, breach risk score calculation and selected action logging using Deluge scripting.
- To train and deploy a Zoho Creator prediction model that assigns a predicted risk category to every submitted breach incident.
- To configure a conditional e-mail notification workflow that alerts the responsible users when a breach incident is predicted to be High risk.
- To provide reports and a dashboard with key performance indicator cards and charts for monitoring privacy operations.
- To test the application, document the defects encountered, and state honestly the limitations of the prototype and of the synthetic training data.

1.4 Scope of the Project

In scope. The application contains six forms (Person_Register, Consent_Records, Rights_Requests, Breach_Incidents, Action_Log and Rule_Configuration), reports on

these forms, and a dashboard with key performance indicator cards and charts. Deluge scripting implements rights request classification, deadline-risk assessment, breach risk score calculation and selected action logging. A prediction model, ConsentLedger_Breach_Risk, is deployed on the Breach_Incidents form, and the ML_High_Risk_Email_Alert workflow sends an e-mail when the predicted category is High. The application is used through a web browser by the application administrator and authorised application users.

Out of scope. The following are not part of the current prototype and are discussed as future enhancements in Chapter 9: production deployment, certification of legal compliance with the DPDP Act, integrations with external systems, Zoho Catalyst or Zoho Analytics, automated statutory breach reporting, scheduled escalation of rights requests, a portal for data principals, a fully configured multi-role access model, a customised native mobile application and an immutable audit trail.

1.5 Justification for a Low-Code Approach

Low-code platforms let developers assemble business applications mainly through visual configuration of data models, forms, reports and workflows, using scripting where custom logic is required [2], [3]. Zoho Creator provides hosting, a database, form and report builders, pages, a workflow engine and the Deluge scripting language in one environment [4], [5]. Consequently, no separate server or hand-coded user interface was required for ConsentLedger.

The approach suits this project for four reasons. First, the problem is dominated by structured data capture, status tracking and notification, which are the core strengths of forms, reports and workflows. Second, the machine-learning component, a breach risk classifier, could be trained and deployed from within the same platform and attached directly to the Breach_Incidents form, so no separate model-serving infrastructure was needed. Third, rules such as the classification of requests can be changed quickly in the builder when a requirement changes. Fourth, user sign-in and hosting are provided by the platform. The limitations of the approach are discussed in Section 8.4.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7

discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements. The appendices contain the feature coverage checklist, course outcome mapping, Deluge source code, user manual, evaluation rubric and viva-voce record sheet.

Throughout the report, features are identified as implemented, partially implemented or not implemented. Details that could not be verified from the application evidence are explicitly marked as requiring confirmation before final submission.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a typical small organisation the privacy-related work is spread across several informal channels. A person's details are kept in a contact spreadsheet, and consent is recorded on a paper form or inferred from the fact that the person supplied their details. When an individual asks to see, correct or erase their data, the request arrives by e-mail or phone, is noted in a register or spreadsheet, and is passed on to whoever is expected to handle it. The date by which a response is needed is remembered rather than calculated. When a security incident occurs, it is reported verbally or by message, its seriousness is judged by the person who receives the report, and the responsible officer is informed by phone or e-mail if the reporter remembers to do so. Figure 2.1 shows this flow.

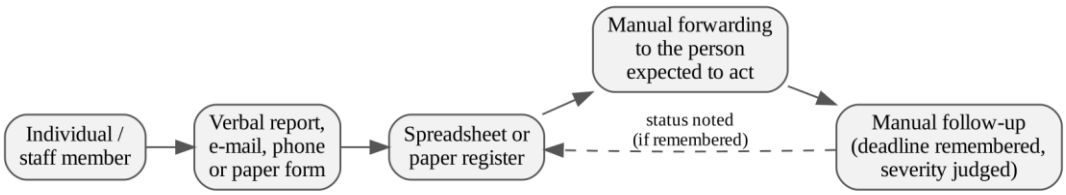


Figure 2.1: As-Is Process Flow of the Existing System

Every hand-over in this flow depends on a person remembering to pass the information on, and no step produces a record that could later be reviewed.

2.2 Limitations of the Existing System

- Personal data, consent records, requests and incidents are stored in different places, so the information is incomplete and often duplicated.
- There is no consistent record of when consent was given or changed, so consent status cannot be verified quickly.
- Rights requests are classified by individual judgement, and the time remaining for each request is not calculated or monitored.
- Incident severity is assessed subjectively, so two officers may rate the same incident differently.

- Escalation of serious incidents depends on a person remembering to notify the right officer.
- No log records which actions were taken, and management has no summary view of consent, requests and incidents.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for ConsentLedger. Low-code platforms reduce the amount of hand-written code needed for many business applications [2], [6]. No effort measurements were taken during this project, so the comparison is qualitative and no development-time figures are claimed.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Separate design and coding of database, back end and front end (not measured for this project)	Forms, reports, pages and workflows configured in one platform (not measured)
Skills required	Programming language, database, web framework, ML tooling and hosting	Form and workflow configuration; Deluge for custom logic
Hosting & infrastructure	Server or cloud setup must be arranged and maintained	Hosted by Zoho; no server set up by the developer
Machine-learning component	Separate model training, packaging, serving and integration with the application	Prediction model trained and deployed inside Zoho Creator and attached to the Breach_Incidents form
Mobile application	Separate mobile development required	Zoho Creator mobile app can open the application (not customised in this project)
Security & access control	Authentication and authorisation coded manually	Sign-in, user fields and permission features provided by the platform
Maintenance & change	Code change, testing and redeployment for each change	Fields, choices, rules and workflows edited in the builder
Customisation limits	Almost unlimited	Limited to what the platform and Deluge allow
Cost	Developer time, server and licence cost	Platform subscription or trial plan Not verified in the current application evidence (confirm plan)

2.4 Literature Survey

Seven sources were studied: three research publications on low-code platforms [2], [6], [7], the DPDP Act [1], Zoho Creator documentation [4], the Deluge reference [5] and the course reference book [3]. Table 2.2 summarises them.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021 [2]	Low-Code Platform	Describes what a low-code platform is and what its main characteristics are.	Gives the definition of low-code used in Chapter 1.
2	A. Sahay et al., 2020 [6]	Supporting the understanding and comparison of low-code development platforms	Proposes a way to compare low-code platforms by their features and shows that platforms differ in capability.	Supports the choice of a platform whose features match the project needs.
3	Y. Luo et al., 2021 [7]	Characteristics and challenges of low-code development: the practitioners' perspective	Studies practitioners' discussions to identify the benefits and the challenges they report with low-code.	Relevant to the limitations discussed in Section 8.4.
4	Government of India, 2023 [1]	The Digital Personal Data Protection Act, 2023	Establishes a framework for processing digital personal data, covering consent, rights of Data Principals and obligations of Data Fiduciaries, including action on personal data breaches.	Provides the privacy context from which the modules of ConsentLedger are drawn. The prototype is not a compliance certification.
5	Zoho Corporation [4]	Zoho Creator Help Documentation	Explains forms, reports, pages, workflows, user management and AI features in Zoho Creator.	Primary reference for the implementation in Chapter 5.
6	Zoho Corporation [5]	Deluge Script Reference Guide	Describes the Deluge scripting language used in Zoho Creator.	Reference for the scripts used for classification, scoring and logging.
7	W. Rosam and O. Kalnichevski [3]	Low-Code Development: A Practical Guide	Course reference book on low-code development.	Background for the low-code concepts used in the report.

2.5 Emerging Trends in Low-Code Development

Three trends in low-code development are relevant to ConsentLedger. The first is citizen development, in which people who understand a business process, and not only professional programmers, build and maintain applications [7]. A privacy operations

tool is a good example, because the person who understands the consent or incident process can adjust forms and rules directly. The second is the embedding of machine learning inside low-code platforms, so that a model can be trained and attached to a form without separate infrastructure. ConsentLedger follows this trend with the ConsentLedger_Breach_Risk prediction model. The third is hyper-automation, in which calculations, record creation and notifications are chained together without manual effort; the combination of the risk score, the predicted category and the conditional e-mail alert applies this idea on a small scale.

The same sources warn that customisation limits and dependence on the platform vendor remain real challenges [7]. These are examined for this project in Section 8.4. Trends such as composable applications and AI-assisted application building were not used in this prototype.

2.6 Proposed System

The proposed system is ConsentLedger, a Zoho Creator application that keeps person records, consent records, rights requests, breach incidents, action logs and rule configuration in six forms. Deluge scripts classify rights requests, assess deadline risk, calculate a risk score for each breach incident and write selected actions to the action log. A deployed prediction model adds a predicted risk category to each incident, and a workflow sends an e-mail alert when that category is High. Reports and a dashboard present the information for monitoring.

Each limitation in Section 2.2 is addressed as follows. Storing all records in one application removes scattered files. The Consent_Records form and the Action_Log give a recorded history of consent creation. Rights request classification and deadline-risk assessment replace judgement-based sorting and memory-based deadlines. The Risk_Score and the predicted risk category make incident assessment repeatable. The conditional e-mail alert reduces dependence on a person remembering to escalate, and the dashboard gives management the summary that was missing. The alert is an aid to escalation and does not replace human investigation or the organisation’s incident-response procedure.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

The stakeholders below are conceptual. They describe who would use or be affected by the system. Which of them have been configured as separate Zoho Creator roles or permission profiles is not established in this report; the only role evidenced by the application configuration is the Application Administrator, and the Assigned_Officer field of Breach_Incidents selects from the Zoho Creator users of the account (Users → All Users).

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Application Administrator	The person who owns and configures the application.	Maintain forms, rules and reports; manage users; monitor the dashboard; view all records.
Authorised Application User	A signed-in Zoho Creator user who has access to the application. Separate custom permission profiles are not established in this report.	Register persons, record consent, enter rights requests and breach incidents, and view reports.
Assigned Officer	A Zoho Creator user selected in the Assigned_Officer field of a breach incident.	Investigate and resolve the incident; record the resolution summary.
Data Principal (conceptual)	The individual whose personal data is recorded. Does not log in; no portal exists.	Appears as a record in Person_Register; their consent and requests are recorded by authorised users.

3.2 Functional Requirements

Each requirement below is testable, and the same identifiers are used in the test cases of Chapter 6.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall store information about each data principal in the Person_Register form.	Administrator, Authorised User	High

FR ID	Requirement	Role(s)	Priority
FR-02	The system shall store consent records and consent lifecycle information in the Consent_Records form.	Administrator, Authorised User	High
FR-03	The system shall record a log entry in Action_Log when a consent record is created.	System	Medium
FR-04	The system shall allow a data principal's request about their personal data to be captured in the Rights_Requests form.	Administrator, Authorised User	High
FR-05	The system shall classify each rights request using Deluge logic on submission.	System	High
FR-06	The system shall assess the deadline risk of each rights request using Deluge logic.	System	High
FR-07	The system shall capture a breach incident with its reference ID, title, description, discovery date and time, severity, status, estimated affected records, assigned officer, response due date and time and resolution summary.	Administrator, Authorised User	High
FR-08	The system shall calculate and store a numerical Risk_Score for a breach incident using its severity and affected-records information.	System	High
FR-09	The system shall populate the Predicted_Risk_Category field of each submitted breach incident using the deployed ConsentLedger_Breach_Risk model.	System	High
FR-10	The system shall send an e-mail alert when a newly submitted breach incident has Predicted_Risk_Category equal to High, and shall not send it for other categories.	System	High
FR-11	The system shall store operational rule configuration in the Rule_Configuration form.	Administrator	Medium
FR-12	The system shall provide reports for viewing records of the forms.	Administrator, Authorised User	Medium
FR-13	The system shall display a dashboard with key performance indicator cards and charts of privacy operations information.	Administrator	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall load within an acceptable time for prototype-scale data. No measurement has been taken.

NFR ID	Category	Requirement
NFR-02	Security	Only signed-in Zoho users shall access the application. A fully configured multi-role access model is not claimed.
NFR-03	Usability	The application shall be usable from a standard web browser with clear form labels and menus.
NFR-04	Availability	The application shall rely on the availability of the Zoho Creator cloud service.
NFR-05	Maintainability	Fields, rules and workflows shall be editable in the Zoho Creator builder without redeploying code.
NFR-06	Auditability	Selected actions, including consent creation, shall be recorded in Action_Log. The log is not claimed to be an immutable audit trail.
NFR-07	Transparency of prediction	The predicted risk category shall be stored beside the rule-based Risk_Score so that users can compare them. It is an advisory output and not a final decision.
NFR-08	Data protection in testing	Testing shall use fictitious or synthetic records and not real personal data.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Education edition)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
AI feature	Zoho Creator prediction model (ConsentLedger_Breach_Risk)
Notification	Zoho Creator Send Email workflow action
Other Zoho services	None configured
Mobile	Not customised
Client hardware	Any computer with a modern browser and an internet connection

3.5 Use Case Model

Figure 3.1 shows the main use cases. The administrator maintains the application and monitors the dashboard. Authorised users register persons, record consent, enter rights requests and breach incidents, and view reports. The assigned officer receives the high-risk alert and handles the incident. The Data Principal is shown as an external actor who

does not use the application directly. Table 3.5 and Table 3.6 describe two key use cases in detail.

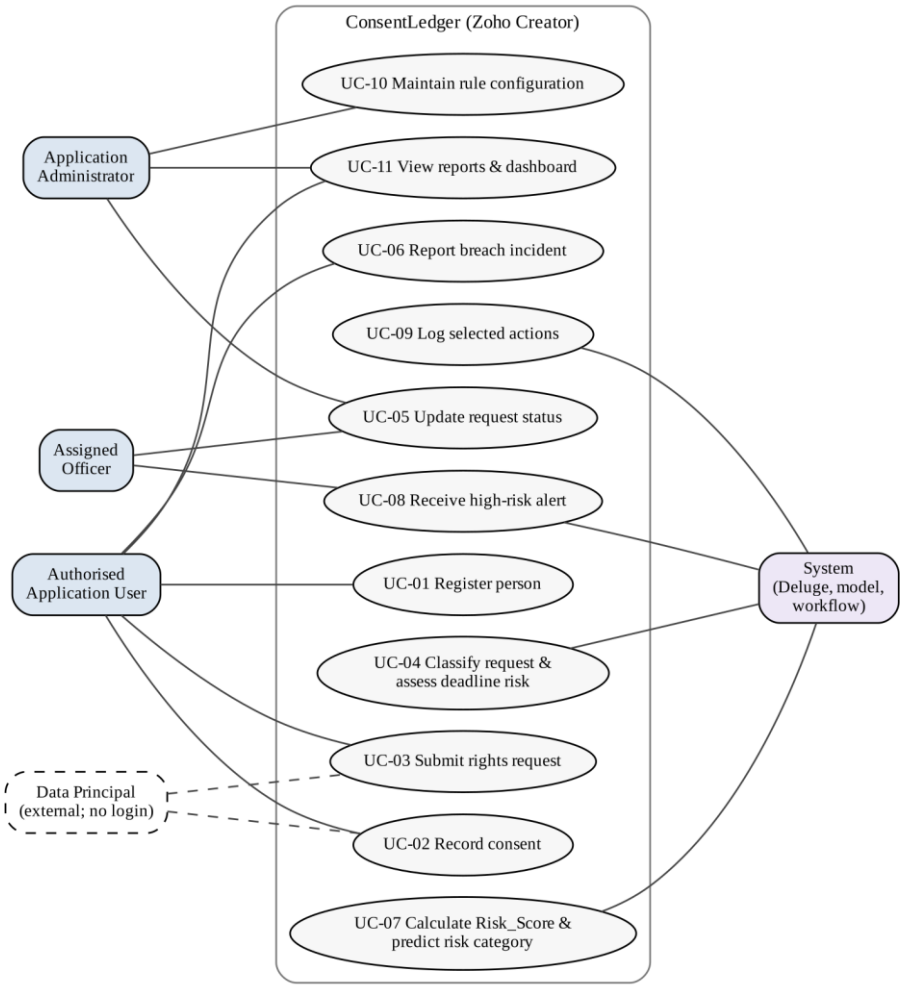


Figure 3.1: Use Case Diagram of ConsentLedger

Table 3.5: Use Case Description – Submit a Breach Incident and Receive a High-Risk Alert

Use Case ID	UC-01
Use Case Name	Submit a breach incident and receive a high-risk alert
Actor(s)	Authorised Application User; Assigned Officer (receives the e-mail alert)
Pre-condition	The user is signed in to Zoho Creator. The prediction model is deployed on Breach_Incidents and the ML_High_Risk_Email_Alert workflow is active.

Main Flow	1. The user opens Breach_Incidents and enters the incident details, including severity, estimated affected records and the assigned officer. 2. The user submits the form. 3. The Risk_Score is calculated. 4. The prediction model stores a value in Predicted_Risk_Category. 5. If the value is High, the workflow sends the alert e-mail with the incident details. 6. The assigned officer investigates the incident.
Alternate Flow	If the predicted category is Medium or Low, no e-mail is sent. The record is stored and remains visible in the reports.
Post-condition	The incident is stored with its Risk_Score and Predicted_Risk_Category. A High prediction has triggered an e-mail.
Related FR	FR-07, FR-08, FR-09, FR-10

Table 3.6: Use Case Description – Submit and Classify a Rights Request

Use Case ID	UC-02
Use Case Name	Submit and classify a rights request
Actor(s)	Authorised Application User
Pre-condition	The data principal exists in Person_Register. The user is signed in.
Main Flow	1. The user opens Rights_Requests and enters the request details on behalf of the data principal. 2. The user submits the form. 3. The Deluge logic classifies the request. 4. The Deluge logic assesses the deadline risk of the request. 5. The classified request is visible in the reports and dashboard.
Alternate Flow	If required information is missing, the user must review and complete the form fields before submission; no additional custom validation rule is claimed.
Post-condition	The request is stored with its classification and deadline-risk assessment. The request is not claimed to be processed or completed automatically.
Related FR	FR-04, FR-05, FR-06

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

Figure 4.1 shows the architecture of ConsentLedger as five conceptual layers. The layers describe how the application is organised inside Zoho Creator. No external service is connected.

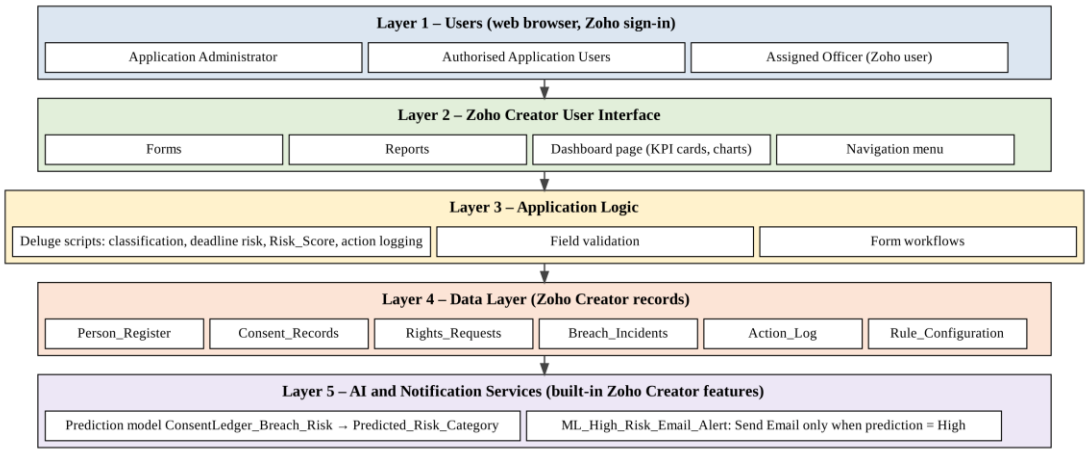


Figure 4.1: System Architecture of ConsentLedger

Table 4.1: Architecture Layers of ConsentLedger

Layer	Name	Components
Layer 1	Users	Application administrator and authorised application users, who reach the application through a web browser after signing in to Zoho.
Layer 2	Zoho Creator User Interface	Forms, reports, dashboard pages and the application navigation menu.
Layer 3	Application Logic	Deluge scripts, validation, calculations (such as the risk score) and workflow automation.
Layer 4	Data Layer	Zoho Creator application records stored in the six forms: Person_Register, Consent_Records, Rights_Requests, Breach_Incidents, Action_Log and Rule_Configuration.
Layer 5	AI and Notification Services	The ConsentLedger_Breach_Risk prediction model and the conditional ML_High_Risk_Email_Alert workflow.

Data flows from top to bottom. A user enters data through a form in Layer 2. Layer 3 validates the input and runs the scripts that classify requests, assess deadline risk and calculate scores. The record is written to the corresponding form in Layer 4. For a breach incident, the prediction model in Layer 5 populates Predicted_Risk_Category,

and the workflow in the same layer sends an e-mail when that value is High. Reports and the dashboard in Layer 2 read the stored records back for monitoring. Zoho Catalyst, Zoho Analytics, external APIs and third-party databases are not part of the implemented architecture and appear only as future enhancements in Section 9.2.

4.2 Data Model

Figure 4.2 shows the entity–relationship diagram of the six forms. In Zoho Creator a relationship between forms is normally created with a lookup field, and a parent–child structure with a subform. In this report a relationship is shown only where it has been confirmed in the application; the remaining relationships are marked for confirmation so that the diagram does not claim a link that was not built.

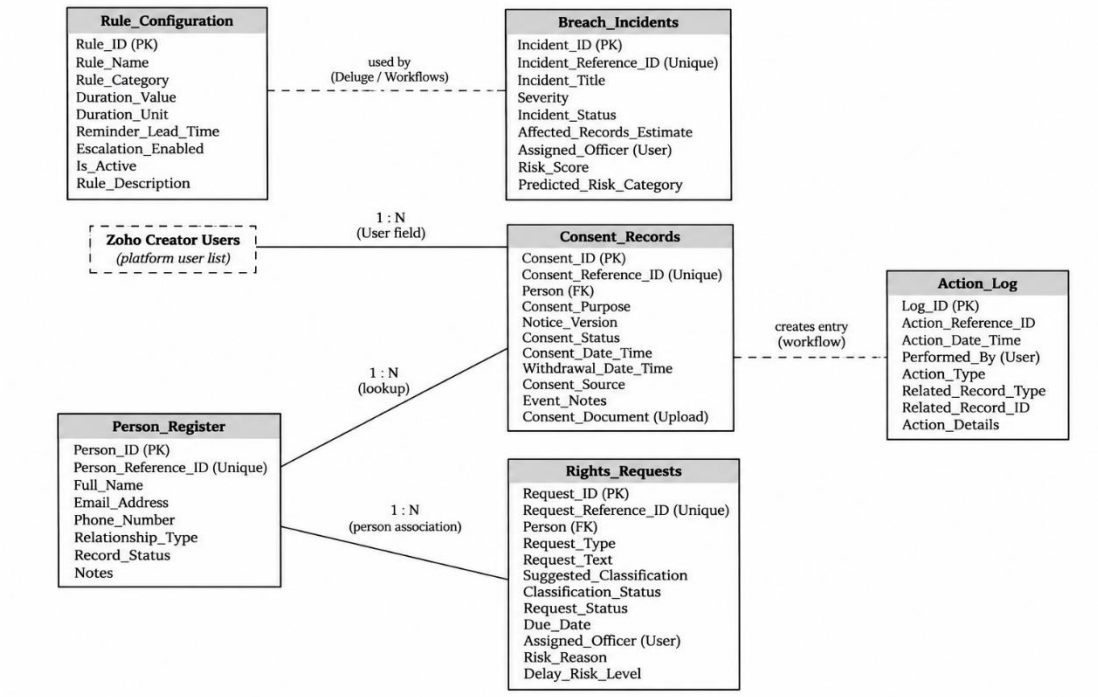


Figure 4.2: Entity–Relationship Diagram of ConsentLedger

Table 4.2: Entities and Relationships

Entity (Form)	Purpose	Key Fields	Relationships
Person_Register	Maintains information about the individuals (data principals) whose data and consent are managed.	Person_Reference_ID, Full_Name, Email_Address, Phone_Number, Relationship_Type, Record_Status, Notes	One person can be associated with multiple Consent_Records and Rights_Requests through lookup fields.

Entity (Form)	Purpose	Key Fields	Relationships
Consent_Records	Maintains consent records and lifecycle details.	Consent_Reference_ID, Person, Purpose, Notice_Version, Consent_Status, Consent_Date_Time, Withdrawal_Date_Time, Source, Event_Notes, Consent_Document, OCR_Review_Status	Each consent record is associated with a person through the Person lookup field. A workflow creates an Action_Log entry when consent is created.
Rights_Requests	Captures requests made by data principals about their personal data.	Request_Reference_ID, Person, Request_Type, Request_Text, AI_Suggested_Type, Classification_Status, Request_Status, Due_Date, Assigned_To, Delay_Risk_Level, Risk_Reason, Request_Date, Completion_Date, Resolution_Notes	Each rights request is associated with a person through the Person lookup field.
Breach_Incidents	Captures personal data breach incidents.	Incident_Reference_ID, Incident_Title, Incident_Description, Discovery_Date_Time, Severity, Incident_Status, Affected_Records_Estimate, Assigned_Officer, Response_Due_Date_Time, Risk_Score, Predicted_Risk_Category, Resolution_Summary	Assigned_Officer is a User field that selects from Zoho Creator Users → All Users. A High-risk prediction triggers the configured e-mail workflow.
Action_Log	Records selected actions of the application, including consent creation.	Action_Reference_ID, Action_Date_Time, Added_User, Action_Type, Related_Record_Type, Related_Record_ID, Action_Details	A workflow creates an Action_Log entry when a consent record is created. Related records are identified using Related_Record_Type and Related_Record_ID.

Entity (Form)	Purpose	Key Fields	Relationships
Rule_Configuration	Stores operational rule configuration.	Rule_Name, Rule_Category, Duration_Value, Duration_Unit, Reminder_Lead_Time, Escalation_Enabled, Is_Active, Rule_Description	No direct lookup relationship with the other forms has been confirmed.

4.3 Form Design

This section gives the field specification of each form. Field names and types are listed only where they have been verified. For the Breach_Incidents form the twelve confirmed fields are listed in Table 4.3. For the other five forms the tables are left ready to be completed from the form builder, and the corresponding screenshots are requested in Chapter 5. Entries marked for confirmation must be completed before submission.

Table 4.3: Field Specification – Breach_Incidents Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Incident Reference ID	Incident_Reference_ID	Single Line	Yes	Unique incident reference; shown in alert.
Incident Title	Incident_Title	Single Line	Yes	Short incident name; shown in alert.
Incident Description	Incident_Description	Multi Line	Yes	Description of the incident.
Discovery Date Time	Discovery_Date_Time	Date-Time	Yes	When incident was discovered.
Severity	Severity	Dropdown	No	Used in risk assessment and prediction.
Incident Status	Incident_Status	Dropdown	No	Current incident state; shown in alert.
Affected Records Estimate	Affected_Records_Estimate	Number	No	Estimated affected records.
Assigned Officer	Assigned_Officer	User	No	Selects from Users → All Users.
Response Due Date Time	Response_Due_Date_Time	Date-Time	No	Target response date/time.
Resolution Summary	Resolution_Summary	Multi Line	No	Summary of resolution.
Risk Score	Risk_Score	Number	No	Rule-based numerical assessment.
Predicted Risk Category	Predicted_Risk_Category	Prediction output	System populated	Output from deployed Zoho Creator model.

The original training dataset contained Incident_Type and Containment_Status in addition to Severity and Affected_Records_Estimate. The final deployed form mapping

must be verified against the current prediction deployment configuration; this report does not assume that fields absent from the form were mapped.

Table 4.4: Field Specification – Person_Register Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Full Name	Full_Name	Name (First/Last)	Yes	Person's name.
Email Address	Email_Address	Email	Yes	Contact email.
Phone Number	Phone_Number	Phone	No	Contact number.
Person Reference ID	Person_Reference_ID	Single Line	No	Reference identifier.
Relationship Type	Relationship_Type	Dropdown	No	Relationship classification.
Record Status	Record_Status	Dropdown	No	Record state.
Notes	Notes	Multi Line	No	Additional information.

Table 4.5: Field Specification – Consent_Records Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Person	Person	Lookup	Yes	Links consent to a registered person.
Purpose	Purpose	Dropdown	Yes	Purpose for processing.
Notice Version	Notice_Version	Single Line	Yes	Version of notice presented.
Consent Status	Consent_Status	Dropdown	Yes	Current consent state.
Consent Date Time	Consent_Date_Time	Date-Time	Yes	Date/time consent was recorded.
Withdrawal Date Time	Withdrawal_Date_Time	Date-Time	No	Date/time consent was withdrawn.
Source	Source	Dropdown	No	Channel through which consent was obtained.
Event Notes	Event_Notes	Multi Line	No	Additional consent event details.
Consent Document	Consent_Document	File Upload	No	Uploaded consent document.
OCR Review Status	OCR_Review_Status	Dropdown	No	Review state for OCR-related processing.
Consent Reference ID	Consent_Reference_ID	Single Line	Yes	Consent reference identifier.
OCR	OCR	OCR-related field	No	OCR automation is not claimed.

Table 4.6: Field Specification – Rights_Requests Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Request Reference ID	Request_Reference_ID	Single Line	No	Unique request reference.
Person	Person	Lookup	Yes	Person associated with request.
Request Type	Request_Type	Dropdown	Yes	Access, Correction, Erasure or Other.

Field Label	Link Name	Field Type	Mandatory	Remarks
Risk Reason	Risk_Reason	Multi Line	No	Explanation for deadline-risk assessment.
Due Date	Due_Date	Date	No	Target response date.
Request Text	Request_Text	Multi Line	Yes	Text used for suggested classification.
AI Suggested Type	AI_Suggested_Type	Dropdown	No	Deluge-generated suggested classification; not a model prediction.
Classification Status	Classification_Status	Dropdown	No	Review status of suggested classification.
Request Status	Request_Status	Dropdown	Yes	Current request state.
Assigned To	Assigned_To	User / Lookup	No	Responsible user; exact field type should be checked.
Resolution Notes	Resolution_Notes	Multi Line	No	Notes on handling and resolution.
Delay Risk Level	Delay_Risk_Level	Dropdown	No	High, Medium, Low or Not Assessed.
Request Date	Request_Date	Date	Yes	Date request was received.
Completion Date	Completion_Date	Date	No	Date request was completed.

Table 4.7: Field Specification – Action_Log Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Action Reference ID	Action_Reference_ID	Single Line	Yes	Action record reference.
Action Date Time	Action_Date_Time	Date-Time	Yes	Date/time of action.
Added User	Added_User	User	No	User recorded by the consent-creation logging workflow (zoho.loginuser).
Action Type	Action_Type	Dropdown	No	Category of action.
Related Record Type	Related_Record_Type	Dropdown	No	Associated form or record category.
Related Record ID	Related_Record_ID	Single Line	No	Identifier of related record.
Action Details	Action_Details	Multi Line	No	Description of the action.

Table 4.8: Field Specification – Rule_Configuration Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Rule Name	Rule_Name	Single Line	Yes	Name of configured rule.
Rule Category	Rule_Category	Dropdown	No	Category of rule.
Duration Value	Duration_Value	Number	No	Configured duration amount.
Duration Unit	Duration_Unit	Dropdown	No	Unit for duration.
Reminder Lead Time	Reminder_Lead_Time	Number	No	Configured reminder lead time.
Escalation Enabled	Escalation_Enabled	Decision / Checkbox	No	Escalation setting.

Field Label	Link Name	Field Type	Mandatory	Remarks
Is Active	Is_Active	Decision / Checkbox	No	Indicates whether rule is active.
Rule Description	Rule_Description	Multi Line	No	Description of the rule.

4.4 Process and Workflow Design

Three processes are designed. The first is the handling of a rights request, the second is the assessment of a breach incident, and the third is the high-risk e-mail notification.

4.4.1 Rights Request Workflow

Figure 4.3 shows the rights request process. An authorised user enters a request in Rights_Requests. On submission a Deluge script classifies the request and a second piece of logic assesses its deadline risk. The classified request then appears in the reports and dashboard. The process classifies and flags requests; it does not process or complete the statutory request automatically.

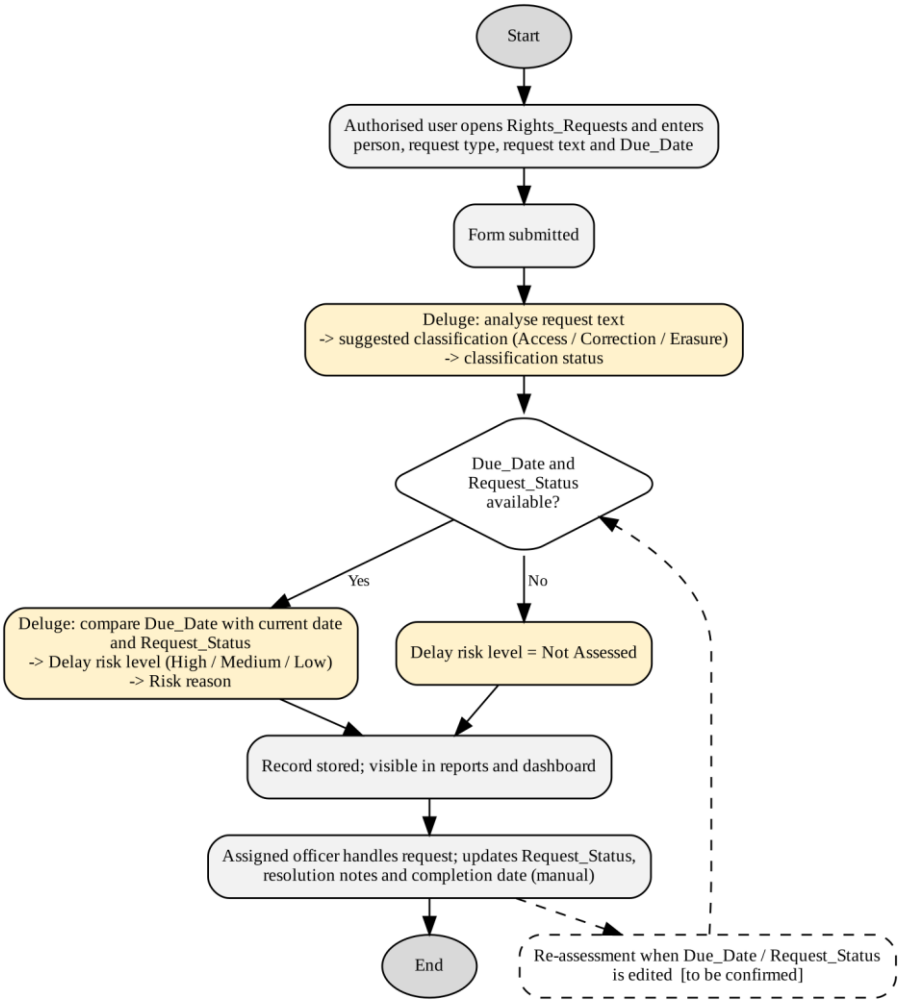


Figure 4.3: Rights Request Workflow

4.4.2 Breach Incident Risk Assessment Workflow

Figure 4.4 shows the breach assessment process. When an incident is submitted, the Risk_Score is calculated by the rule-based logic from the severity and affected-records information. The deployed prediction model independently stores Predicted_Risk_Category. The two outputs are kept together on the record so that a user can read the rule-based score and the model prediction side by side.

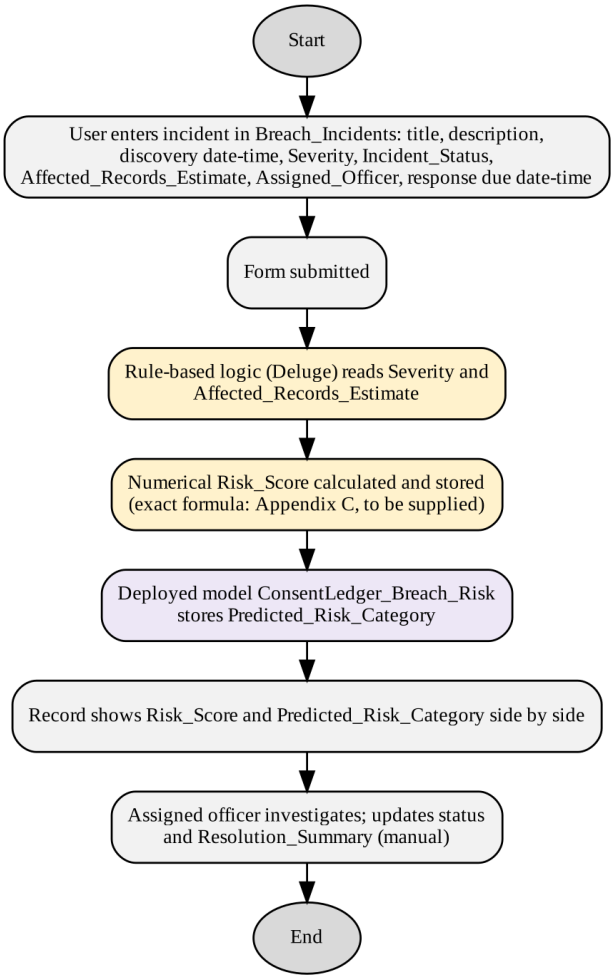


Figure 4.4: Breach Incident Risk Assessment Workflow

4.4.3 E-mail Notification Workflow

Figure 4.5 shows the notification process. The ML_High_Risk_Email_Alert workflow runs on the Breach_Incidents form when a record is created and the form is submitted successfully. A condition checks whether Predicted_Risk_Category equals High. Only then is the Send Email action executed. The data flow is: breach incident input → prediction model → predicted risk category → conditional notification workflow.

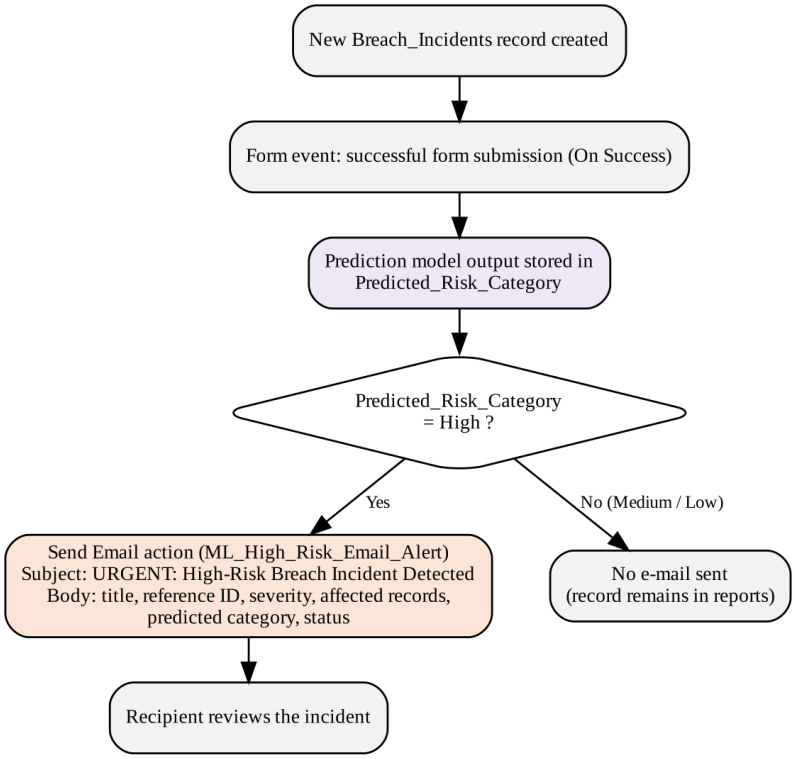


Figure 4.5: E-mail Notification Workflow (ML_High_Risk_Email_Alert)

4.5 User Interface and Navigation Design

The application menu groups the six forms and their reports, with the dashboard as the monitoring page. Figure 4.6 is intended to show the navigation structure together with a wireframe of the dashboard layout, which holds the key performance indicator cards and charts. The exact menu sections, report names and dashboard components must be taken from the live application when the diagram is prepared; none of them is invented here.

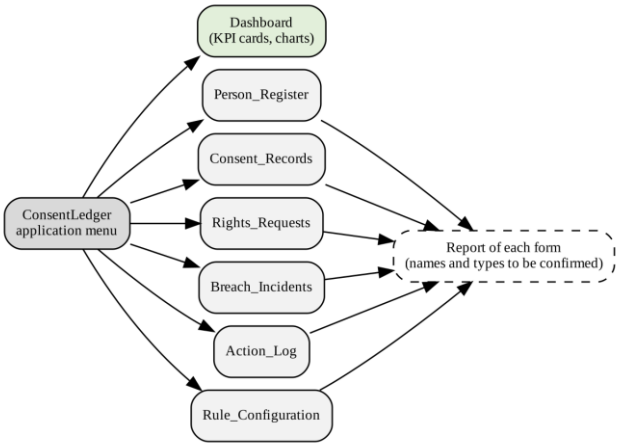


Figure 4.6: Application Navigation Structure and Dashboard Wireframe

4.6 Access Control Design

Zoho Creator provides sign-in, user management, roles and permission-profile capabilities. For ConsentLedger, the application administrator is the owner and configures the application. Separate custom permission profiles and record-level access criteria for other users have not been established. Table 4.9 therefore records the access model as not established rather than assigning unverified permissions.

Table 4.9: Role–Permission Matrix (A = Add, E = Edit, V = View, D = Delete)

Component	Application Administrator	Authorised Application User
Form: Person_Register	Not established in the submitted application evidence	Not established in the submitted application evidence
Form: Consent_Records	Not established in the submitted application evidence	Not established in the submitted application evidence
Form: Rights_Requests	Not established in the submitted application evidence	Not established in the submitted application evidence
Form: Breach_Incidents	Not established in the submitted application evidence	Not established in the submitted application evidence
Form: Action_Log	Not established in the submitted application evidence	Not established in the submitted application evidence
Form: Rule_Configuration	Not established in the submitted application evidence	Not established in the submitted application evidence
Reports	Not established in the submitted application evidence	Not established in the submitted application evidence
Page: Dashboard	Not established in the submitted application evidence	Not established in the submitted application evidence

CHAPTER 5

IMPLEMENTATION

This chapter explains each implemented feature of ConsentLedger: its purpose, its configuration, how it operates and the evidence that supports it. Screenshots are requested at the places where the live application must be shown. Sections for features that were not implemented are retained with the reason.

5.1 Application Setup

The application was created in Zoho Creator under the name ConsentLedger using the Education edition. Zoho Creator hosts the application, so no separate server was configured. Scripting is done in Deluge, and development followed the low-code approach described in Chapter 1. The navigation menu groups the six forms and their reports and provides access to the dashboard. Figure 5.1 shows the application home and navigation sections.

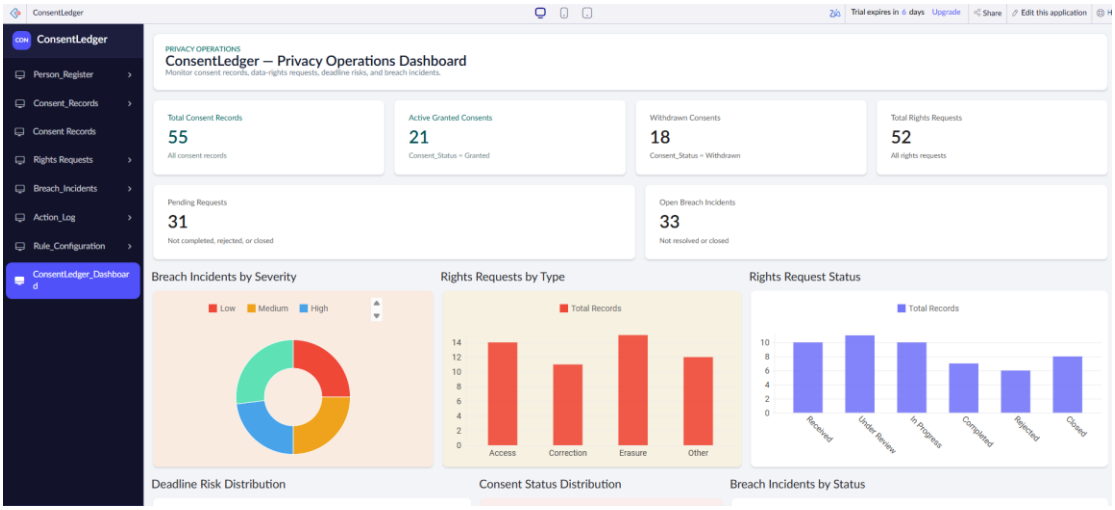


Figure 5.1: Application Home and Navigation Sections of ConsentLedger

5.2 Forms and Data Capture

Six forms capture the data of the system. Fixed-value inputs should be drop-downs or similar choice fields so that users select rather than type, which keeps records consistent. The purpose and role of each form is described below.

5.2.1 Person_Register

Person_Register maintains information about the individuals whose personal data and consent records are managed by the application. It is the starting point of the workflow:

a person is registered before consent or a request is associated with them. The fields of this form are documented in Table 4.4.

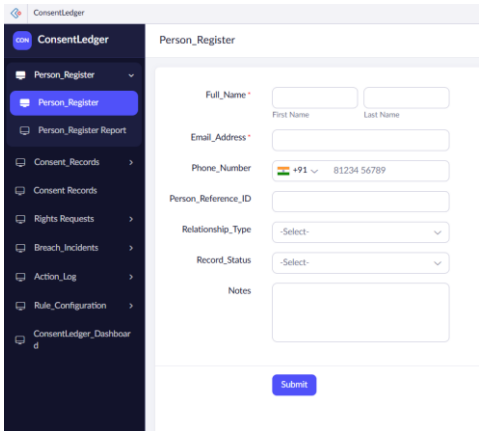


Figure 5.2: Person_Register Form in Zoho Creator

5.2.2 Consent_Records

Consent_Records maintains consent-related records, including consent information and lifecycle details. Keeping the records in a single form gives the organisation one place to look up the consent position instead of searching several files. Creation of a consent record is written to the Action_Log through the logging logic described in Section 5.5, so that there is a record that the consent entry was made.

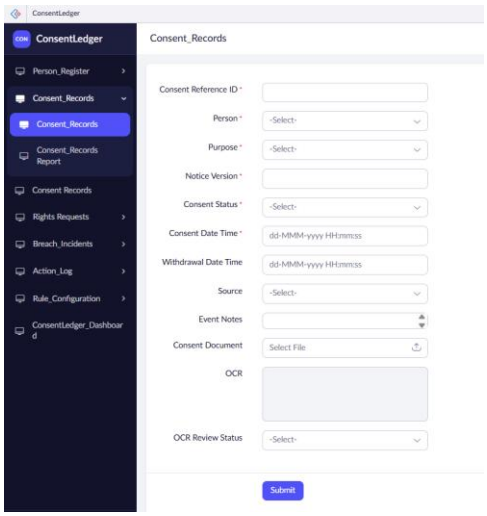


Figure 5.3: Consent_Records Form in Zoho Creator

5.2.3 Rights_Requests

Rights_Requests captures requests made by data principals concerning their personal data. On submission the application applies Deluge-based request classification and deadline-risk assessment (Section 5.5). This reduces dependence on manual sorting and

on remembering deadlines. It does not mean that a statutory request is processed or completed automatically; the response itself remains a human task.

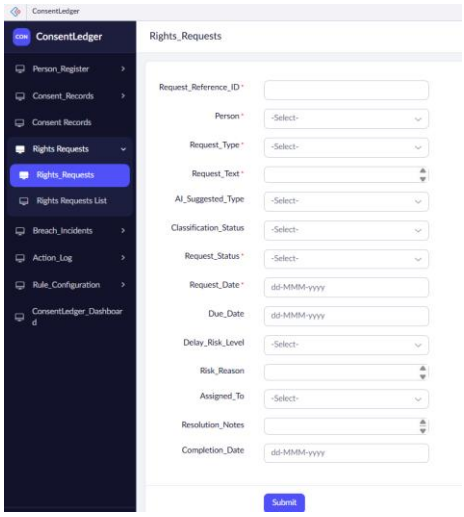


Figure 5.4: Rights_Requests Form in Zoho Creator

5.2.4 Breach_Incidents

Breach_Incidents captures personal data breach incidents. Its confirmed fields are listed in Table 4.3. Assigned_Officer is a Zoho Creator User field configured to select from Users → All Users, so an incident is always assigned to an existing user of the account and not to free text. Risk_Score is a Number field that holds the rule-based assessment. Predicted_Risk_Category is created through the prediction-model deployment process and is populated when a breach record is submitted.

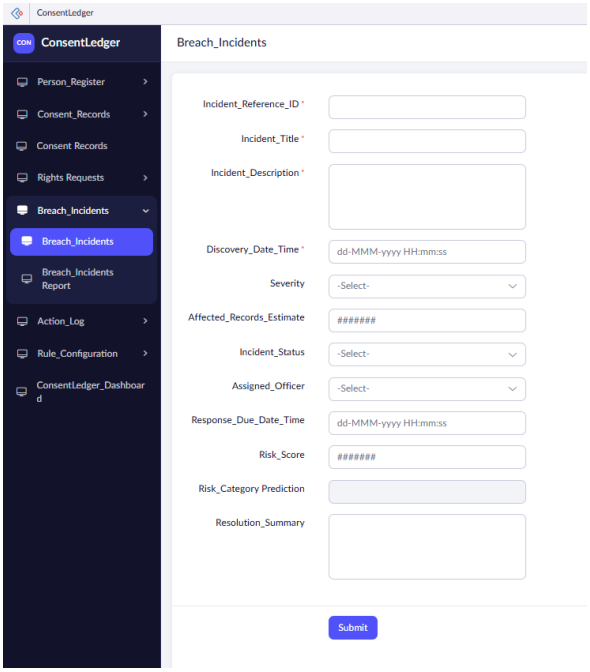


Figure 5.5: Breach_Incidents Form in Zoho Creator

5.2.5 Action_Log

Action_Log maintains action-related records associated with the operation of the application. It records selected actions, including the creation of consent records. It is a convenience log of chosen events. It is not claimed to be a complete or immutable audit trail, because records in a Zoho Creator form can be edited or deleted by users who have permission.

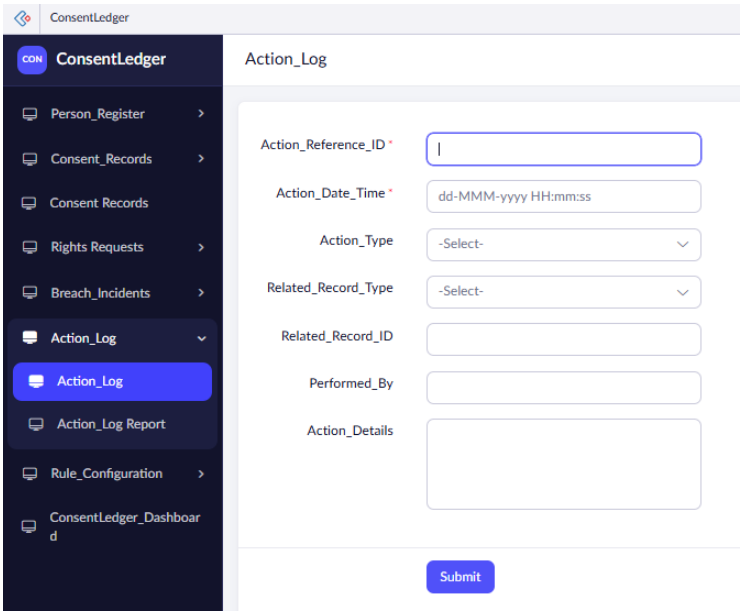


Figure 5.6: Action_Log Form in Zoho Creator

5.2.6 Rule_Configuration

Rule_Configuration stores rule-related information for administrative reference. The current Deluge scripts use fixed logic and do not dynamically retrieve their thresholds or values from this form. Its fields are documented in Table 4.8.

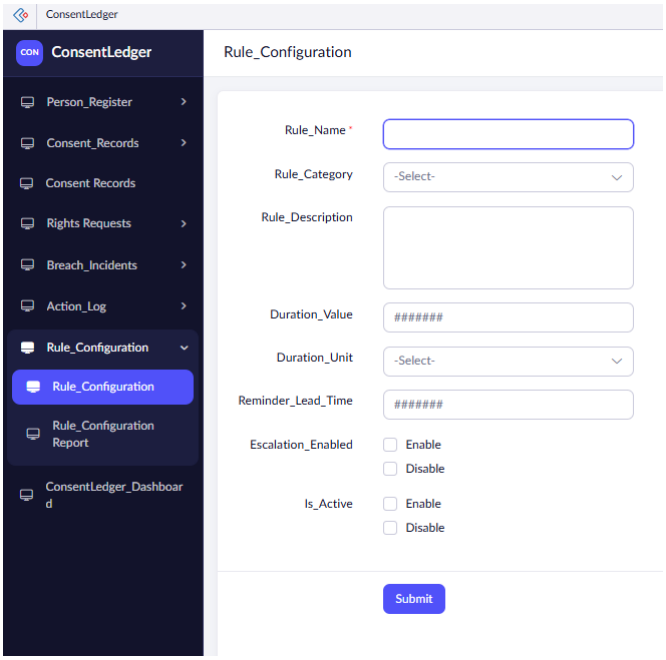


Figure 5.7: Rule_Configuration Form in Zoho Creator

5.3 Reports

Zoho Creator reports are used to view and search the records of the forms. A report is available for the records of each form Exact report names and types to be verified in the application. The reports give the administrator and authorised users a way to review registered persons, consent records, requests, incidents and log entries. Filtering, grouping, sorting and custom actions are to be described where they were configured. Table 5.1 is to be completed from the application.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Person_Register Report	List	Person_Register	Default list view; custom filters not confirmed	None confirmed	Signed-in users with application access
Consent_Records Report	List	Consent_Records	Default list view; custom filters not confirmed	None confirmed	Signed-in users with application access

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Consent Records List	List	Consent_Records	Default list view; custom filters not confirmed	None confirmed	Signed-in users with application access
Rights Requests List	List	Rights_Requests	Default list view; custom filters not confirmed	None confirmed	Signed-in users with application access
Breach_Incidents Report	List	Breach_Incidents	Default list view; custom filters not confirmed	None confirmed	Signed-in users with application access
Action_Log Report	List	Action_Log	Default list view; custom filters not confirmed	None confirmed	Signed-in users with application access
Rule Configuration Report	List	Rule_Configuration	Default list view; custom filters not confirmed	None confirmed	Signed-in users with application access

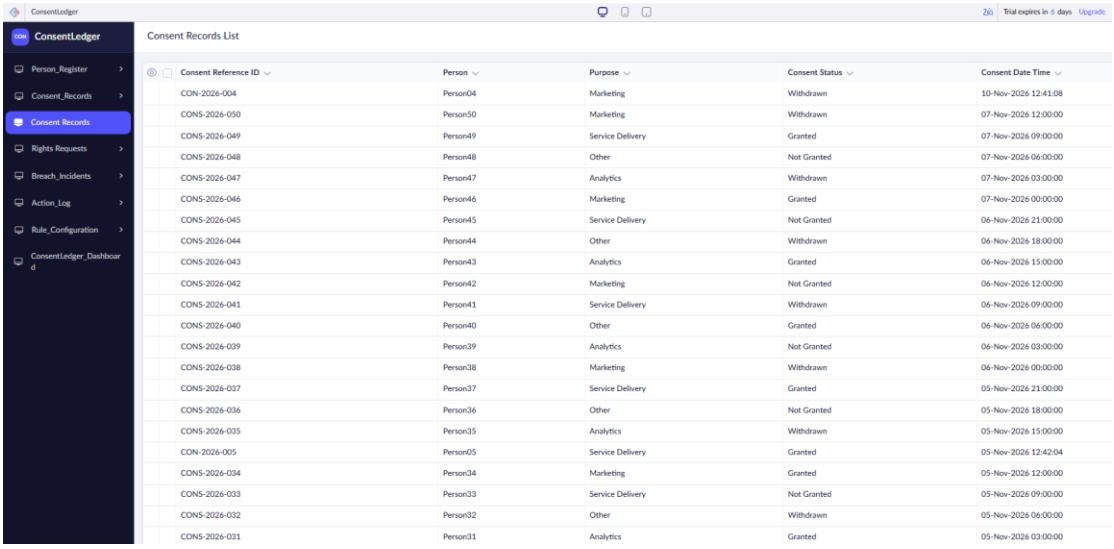


Figure 5.8: Reports in ConsentLedger

5.4 Pages and Dashboards

The dashboard is a Zoho Creator page that presents privacy operations information. It contains key performance indicator cards and charts. Reporting areas that the dashboard is intended to cover are person registration information, consent records, rights request information, breach incident information, risk classification information and action-related records. The purpose of the page is to let the administrator see the overall position without opening each form. The exact cards, charts and values must be taken from the application; no counts are quoted in this report.

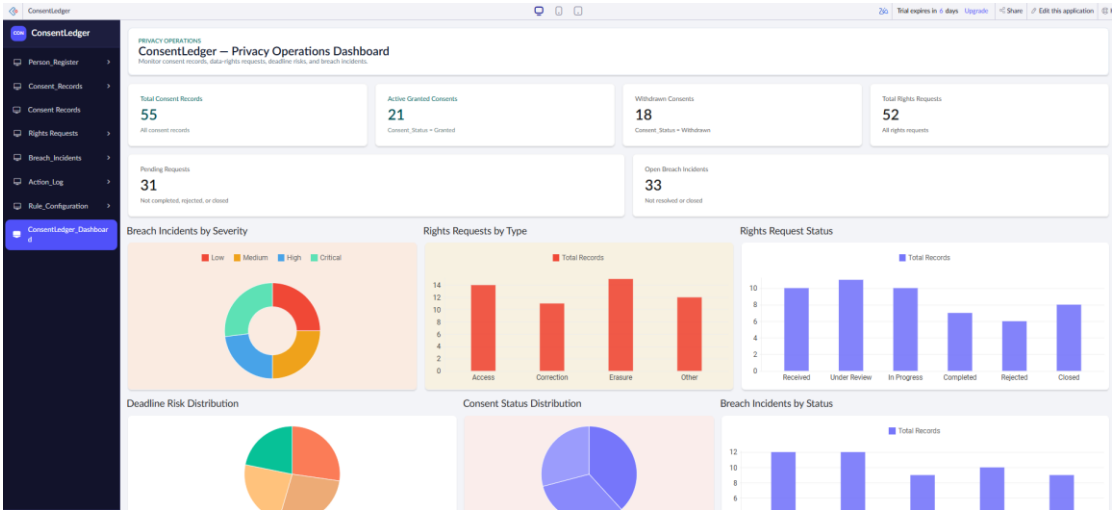


Figure 5.9: ConsentLedger Dashboard with KPI Cards and Charts

5.5 Deluge Scripting

ConsentLedger uses Deluge scripting to implement application-level business logic [5]. Deluge connects the forms, record operations, calculations and workflows. When a record is submitted, a script can read the values that were entered, compute a result, write the result into a field of the same record and add a record to another form such as Action_Log. Table 5.2 lists the automation areas implemented in Deluge. The names of scripts, the events on which they run and the exact rules are to be taken from the application and from Appendix C. No function name, formula or field reference is quoted here that has not been verified.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Rights_Requests → Created; User Input of Request_Text	Classify a rights request	String conversion, lowercase text, contains checks, conditional branching and field assignment
DS-02	Rights_Requests → Created; Successful form submission; Edited on Due_Date and Request_Status	Assess the deadline risk of a request	Null checks, current date, date difference, conditional branching and field assignment
DS-03	Breach_Incidents → Created or Edited; field-event trigger as configured	Calculate Risk_Score from severity and affected-records information	Input field handling, conditional logic and numeric assignment; exact formula must match source code
DS-04	Consent_Records → Created; Successful form submission	Log consent creation in Action_Log	Record ID handling, date-time, string construction, insert into form and user assignment

Rights request classification and deadline-risk assessment (DS-01, DS-02). The classification assigns a category to each request when it is submitted, and the deadline-risk assessment indicates how much risk there is that the request will miss its deadline. Both replace manual judgement with a repeatable rule. No scheduled escalation is claimed; the assessment is evaluated when the record is processed.

Breach risk score (DS-03). The Risk_Score field stores a numerical risk assessment based on incident severity and the estimated number of affected records. The score assigns 1, 3, 6 and 8 points for Low, Medium, High and Critical severity respectively. It adds 2 points when at least 100 records are affected, or 1 point when at least 10 records are affected, and caps the final score at 10. The complete Deluge script is provided in Appendix C.

Action logging (DS-04). When a consent record is created, a log entry is added to Action_Log. Other selected-action logging has not been implemented or verified.

Code 5.1 – DS-01: Rights Request Classification

```
// DS-01: Rights request classification
requestText = ifnull(input.Request_Text,"").toLowerCase();
suggestedType = "Other";

if(requestText.contains("access") || requestText.contains("copy")
|| requestText.contains("provide my data") ||
requestText.contains("personal information"))
{
    suggestedType = "Access";
}
else if(requestText.contains("correct") ||
requestText.contains("update") ||
requestText.contains("inaccurate"))
{
    suggestedType = "Correction";
}
else if(requestText.contains("erase") ||
requestText.contains("delete") || requestText.contains("remove my
data"))
{
    suggestedType = "Erasure";
}

input.AI_Suggested_Type = suggestedType;
input.Classification_Status = "Pending Review";
```

Code 5.2 – DS-02: Deadline-Risk Assessment

```
// DS-02: Rights request deadline-risk assessment
if(input.Request_Status != "Completed" && input.Request_Status !=
"Closed" && input.Request_Status != "Rejected")
{
    if(input.Due_Date == null)
    {
        input.Delay_Risk_Level = "Not Assessed";
        input.Risk_Reason = "Due date is missing; review
required.";
    }
    else
    {
        daysLeft = daysBetween(zoho.currentdate,input.Due_Date);

        if(daysLeft < 0)
        {
```

```
            input.Delay_Risk_Level = "High";
            input.Risk_Reason = "Request is overdue.";
        }
        else if(daysLeft <= 2)
        {
            input.Delay_Risk_Level = "Medium";
            input.Risk_Reason = "Due date is within 2 days.";
        }
        else
        {
            input.Delay_Risk_Level = "Low";
            input.Risk_Reason = "More than 2 days remain before
the due date.";
        }
    }
}
else
{
    input.Delay_Risk_Level = "Not Assessed";
    input.Risk_Reason = "Request is closed or completed; no
pending deadline risk.";
}
```



Figure 5.10: Deluge Script Editor with a ConsentLedger Script

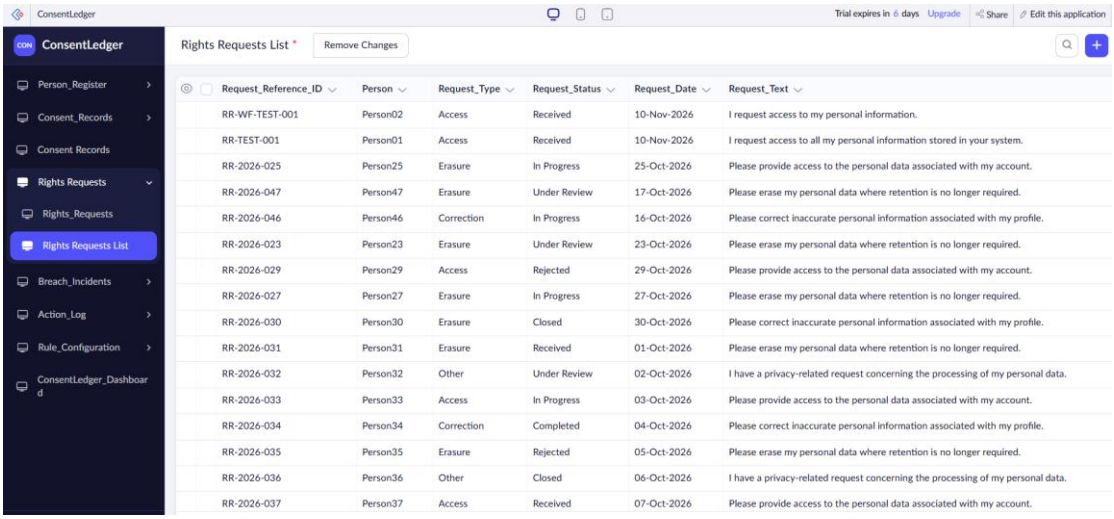


Figure 5.11: Rights Request Classification and Deadline-Risk Result on a Submitted Record

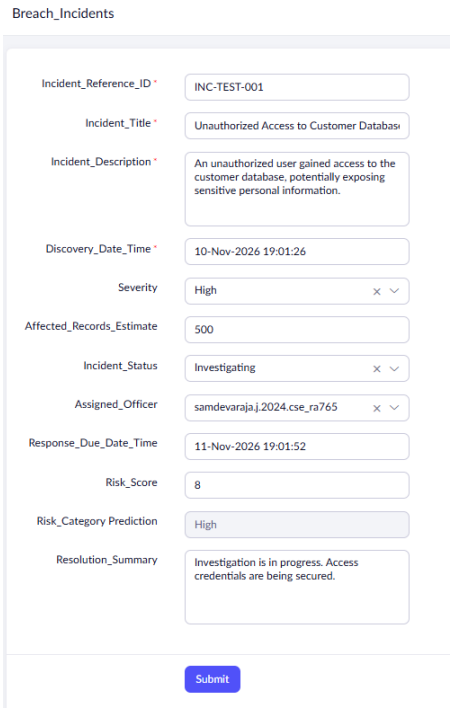


Figure 5.12: Risk_Score Calculated for a Submitted Breach Incident

5.6 Workflows and Automation

Workflows in Zoho Creator execute actions when configured form events occur. The implemented automations are Classify_Rights_Request (Rights_Requests: Created → User Input of Request_Text), Assess_Request_Deadline_Risk (Created → Successful form submission), Update_Deadline_Risk_On_DueDate (Edited → Update of Due_Date), Update_Deadline_Risk_On_Status (Edited → Update of Request_Status), the Breach_Incidents risk-score calculation (DS-03), consent-creation logging on Consent_Records (DS-04), and ML_High_Risk_Email_Alert (Breach_Incidents:

Created → Successful form submission). The email workflow sends an e-mail only when Predicted_Risk_Category equals High. No scheduled, approval, Blueprint or batch workflow is implemented.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Classify_Rights_Request	Form workflow	Rights_Requests – Created; User Input of Request_Text	No additional condition	Deluge classification; sets AI_Suggested_Type and Classification_Status	DS-01
Assess_Request_Deadline_Risk	Form workflow	Rights_Requests – Created; Successful form submission	No additional condition	Calculates Delay_Risk_Level and Risk_Reason	DS-02
Update_Deadline_Risk_On_DueDate	Form workflow	Rights_Requests – Edited; Update of Due_Date	No additional condition	Reassesses deadline risk when Due_Date changes	DS-02
Update_Deadline_Risk_On_Status	Form workflow	Rights_Requests – Edited; Update of Request_Status	No additional condition	Reassesses deadline risk when Request_Status changes	DS-02
ML_High_Risk_Email_Alert	Form workflow	Breach_Incidents – Created; Successful form submission (On Success)	Predicted_Risk_Category equals High	Send Email	–
Breach risk-score calculation	Form workflow	Breach_Incidents – form event (verify exact event in live configuration)	No additional condition stated	Calculates Risk_Score from Severity and Affected_Records_Estimate	DS-03
Consent creation logging	Form workflow	Consent_Records – Created; Successful form submission	No additional condition	Inserts a Consent Created entry into Action_Log	DS-04

5.6.1 ML_High_Risk_Email_Alert

The workflow ML_High_Risk_Email_Alert is attached to Breach_Incidents. Its record event is Created and its form event is successful form submission (On Success), so it runs after a new incident is saved. The condition is Predicted_Risk_Category equals High. When true, the Send Email action sends an alert containing the incident title, reference ID, severity, estimated affected records, predicted risk category and incident status. No notification is sent for Medium or Low predictions.

No notification is configured for Medium or Low predictions. The e-mail is an escalation aid. It does not replace human investigation or the organisation's incident-response procedure. The workflow was tested and the e-mail was received.

During configuration the Send Email action initially returned an error, because Zoho Creator requires the From address to belong to an administrator, a logged-in user or a verified user. The error was removed by selecting a sender address that Zoho Creator recognises. Private e-mail addresses are not reproduced in this report.

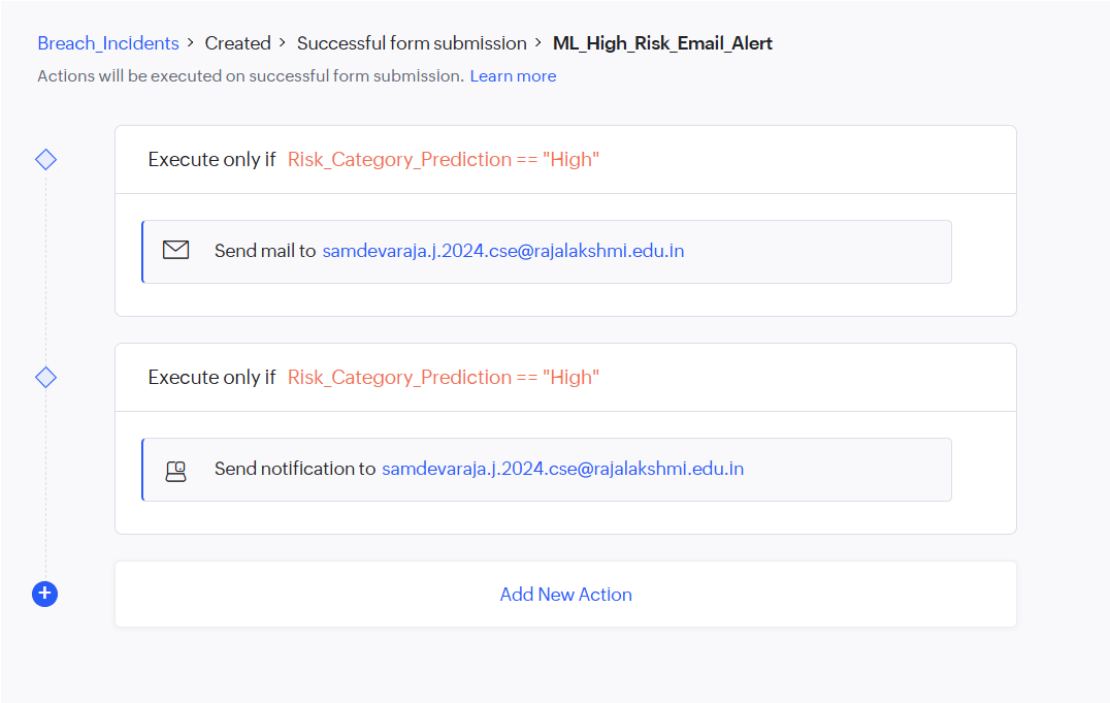


Figure 5.13: ML_High_Risk_Email_Alert Workflow Configuration

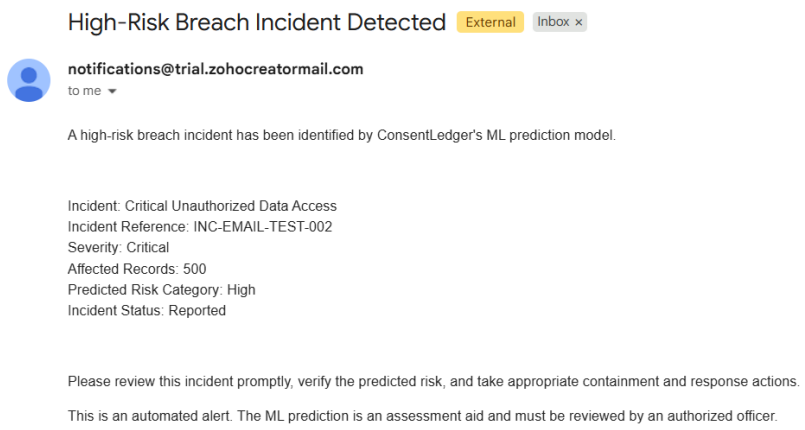


Figure 5.14: High-Risk E-mail Notification Received

5.7 Integrations

No integration with another Zoho service or an external system is implemented. The e-mail alert uses the built-in Send Email action of Zoho Creator, and the prediction model is a Zoho Creator feature. No connection, invokeurl call, webhook or Creator REST API is used, so Table 5.4 has no entries. Zoho Catalyst, Zoho Analytics and integration with external incident management systems are proposed enhancements in Section 9.2.

Table 5.4: External Integrations and Zoho Creator Platform Services

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
External integrations	None configured	Not applicable	Not applicable	No external or cross-application data exchange is implemented.
Zoho Creator Prediction Model	ConsentLedger_Breach_Risk	Built-in prediction model deployment	Zoho Creator application access	Processes breach incident input fields and generate a predicted risk category.
Zoho Creator Deluge	Built-in Deluge scripting environment	Form workflows and event-based execution	Zoho Creator application access	Automates rights request classification, deadline-risk assessment, risk-score calculation and consent creation logging.

5.8 User Management and Role-Based Access

Access to ConsentLedger requires a signed-in Zoho Creator user. The Assigned_Officer field of Breach_Incidents selects from Users → All Users, so officers are selected from the users of the account. A fully configured multi-role access model, with separate profiles and record-level criteria, is not claimed. A portal URL exists, but public/customer portal access and data-principal login have not been verified; therefore, the application is not described as providing operational data-principal portal access. More comprehensive role-based access control is listed as a future enhancement.

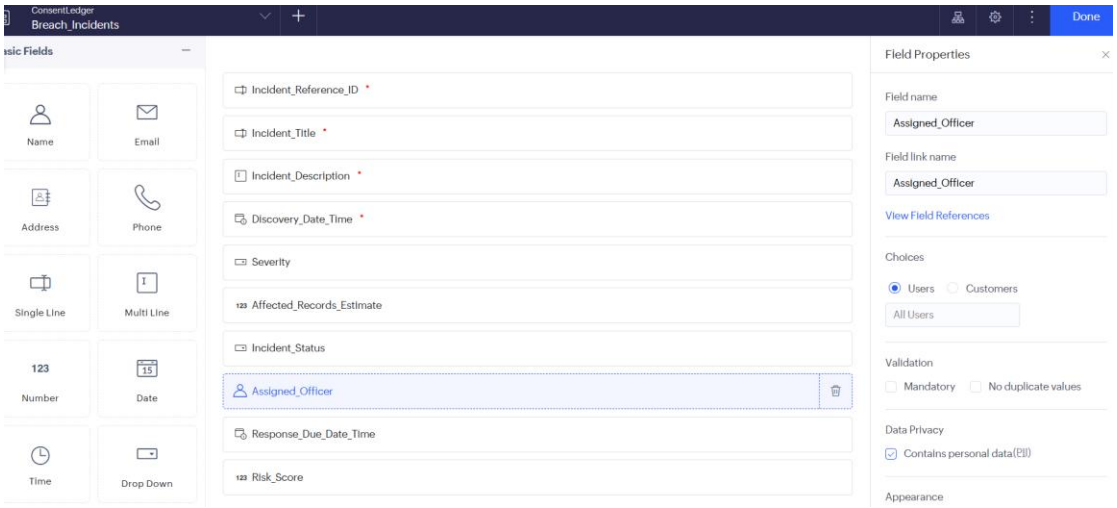


Figure 5.15: Assigned_Officer User Field Configuration

5.9 AI and Analytics Features

5.9.1 Rule-Based Risk Assessment

The application contains a rule-based risk assessment mechanism. The Risk_Score field stores the numerical assessment using severity and the estimated number of affected records. Low, Medium, High and Critical severity contribute 1, 3, 6 and 8 points respectively. The calculation adds 2 points for at least 100 affected records or 1 point for at least 10 affected records, then caps the score at 10. The Deluge implementation is provided as DS-03 in Appendix C.

5.9.2 Zoho Creator Prediction Model

A separate Zoho Creator prediction model named ConsentLedger_Breach_Risk was trained and deployed. The initial training dataset contained 100 synthetic records with the columns Severity, Affected_Records_Estimate, Incident_Type, Containment_Status and Risk_Category. The target column was Risk_Category, and the original model used the other four columns as predictors.

A model test was carried out in Zoho Creator with the following input: Affected Records 75, Containment Status Not Contained, Incident Type Unauthorized Access and Severity High. The model returned a Predicted Risk Category of High with a displayed confidence score of 72.20. The training interface displayed 100% accuracy on the synthetic training dataset.

Table 5.5: Prediction Model Summary

Item	Detail
Model name	ConsentLedger_Breach_Risk
Training data	100 synthetic records
Columns	Severity, Affected_Records_Estimate, Incident_Type, Containment_Status, Risk_Category
Target column	Risk_Category
Predictor columns (original model)	Severity, Affected_Records_Estimate, Incident_Type, Containment_Status
Test input	Affected Records: 75; Containment Status: Not Contained; Incident Type: Unauthorized Access; Severity: High
Test output	Predicted Risk Category: High; displayed confidence score: 72.20
Accuracy shown during training	100% on the synthetic training dataset
Deployment	Published and deployed on the Breach_Incidents form; output field Predicted_Risk_Category

Important qualification. The dataset is synthetic and its labels were generated using illustrative rules. The 100% accuracy shown in the training interface therefore reflects how well the model reproduced those rules on the data it was trained on. It is not evidence of real-world predictive accuracy. The displayed confidence score of 72.20 is the model's own output for one input and must not be read as a validated probability that harm will occur. The model demonstrates the prediction workflow in an academic prototype. Before any operational use it would have to be validated independently with representative, real-world breach data. A model trained on rule-generated labels cannot learn patterns that the rules did not contain, and it inherits every simplification of the rules.

5.9.3 Prediction Model Deployment

The model was published and deployed into ConsentLedger and associated with the Breach_Incidents form. The output field Predicted_Risk_Category is populated in submitted breach records. The data flow is: breach incident input → prediction model → predicted risk category → conditional notification workflow. During setup the model was initially unavailable in the account because of a model reference and configuration issue; this is recorded as a defect in Section 6.4. After the model was available and deployed, test submissions populated the prediction field.

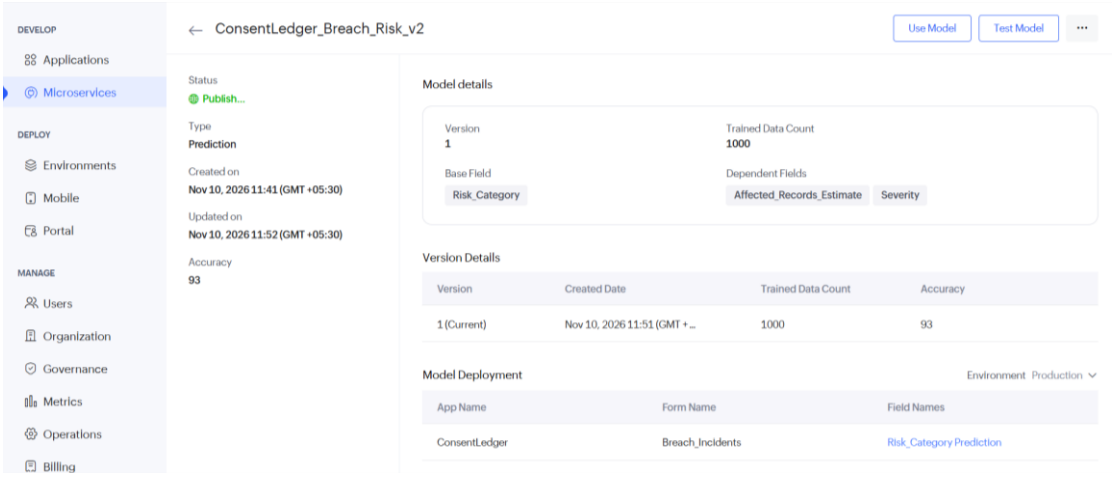


Figure 5.16: ConsentLedger_Breach_Risk Prediction Model Configuration

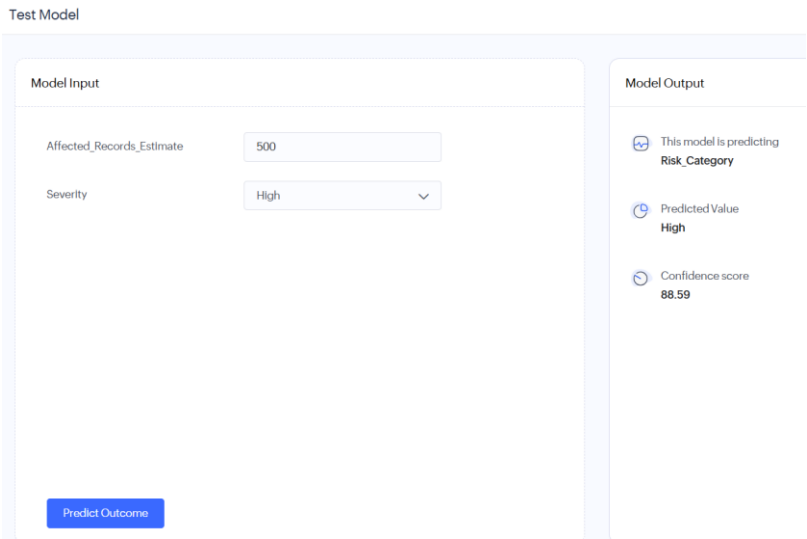


Figure 5.17: Prediction Model Test Output (High, 72.20)

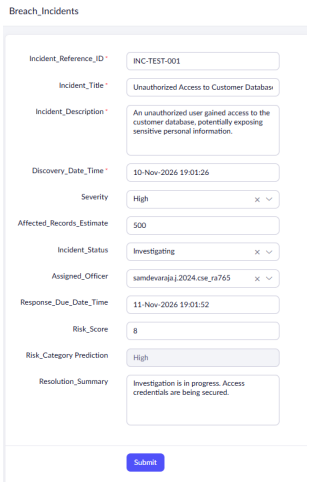


Figure 5.18: Predicted_Risk_Category Populated in a Submitted Breach Record

5.9.4 Analytics

Charts and key performance indicator cards on the dashboard provide the analytics of the application (Section 5.4). Zoho Analytics is not connected. The prediction model was configured by the student in Zoho Creator; its output was checked against the test input given above and against submitted records.

5.10 Pro-Code Extension

No pro-code extension, such as Zoho Catalyst, custom widgets or the Creator REST API, was used. The requirements of this prototype, namely data capture, rule-based scoring, model prediction and notification, were met by forms, Deluge and the built-in model and workflow features, so pro-code was not necessary. In future, a Catalyst service could be used for advanced backend processing or for retraining the model on validated data.

5.11 Mobile and Portal Access

The application is used through a web browser. A customised mobile layout and a portal for data principals were not configured in this academic prototype, because the prototype is operated by the administrator and authorised users and the scope focused on the privacy operations workflow. Zoho Creator applications can be opened in the Zoho Creator mobile app, and a portal for data principals is a possible future enhancement.

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was planned at four levels. Unit testing checks each form for field behaviour and each script for its result. Integration testing checks that the scripts, the prediction model and the workflow work together across forms, for example that a submitted breach incident receives a Predicted_Risk_Category and that the alert workflow reacts to it. Role-based testing signs in as each configured user and checks access. User acceptance testing collects feedback from users who try the application in their role.

This report distinguishes tests with recorded execution evidence from test cases that remain unexecuted. Four tests are recorded as executed: the prediction model test in Zoho Creator, population of Predicted_Risk_Category on submitted breach records, the high-risk e-mail notification test, and correction of the sender-address validation error. These four tests are marked Pass in Table 6.1. The remaining test cases are marked Not executed; their inputs and expected results are documented for future testing. Test data must be fictitious.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a person	Open Person_Register; enter details of a fictitious person; submit	Record saved and visible in the Person_Register report	Person record saved and displayed in Person_Register report.	Pass
TC-02	FR-02	Submit a consent record	Open Consent_Records; select the registered person; enter consent details; submit	Record saved and visible in the Consent_Records report	Consent record saved and displayed in Consent_Records report.	Pass
TC-03	FR-03	Log consent creation	Create a consent record as in TC-02; open Action_Log	A log entry for the consent creation exists	Consent creation entry successfully generated in Action_Log.	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-04	FR-04	Submit a rights request	Open Rights_Requests; enter a request from the registered person; submit	Record saved and visible in the Rights_Requests report	Rights request saved and displayed in Rights_Requests report.	Pass
TC-05	FR-05	Classify a rights request	Submit a request asking for a copy of the person's data	A classification value is assigned according to the configured logic	Request type automatically classified based on the entered request text.	Pass
TC-06	FR-06	Assess deadline risk	Submit one request with a near due date and one with a distant due date	The near-deadline request is assessed with a higher deadline risk than the distant one	Deadline risk level and risk reason correctly calculated based on the due date.	Pass
TC-07	FR-07	Submit a breach incident	Enter title, description, discovery date, severity High, estimate 75, an assigned officer, due date; submit	Record saved with all entered values	Breach incident saved with the entered incident details.	Pass
TC-08	FR-07	Assigned_Officer is a user	Open the Assigned_Officer field	Users of the account are listed (Users → All Users)	Assigned officer successfully selected from the available application users.	Pass
TC-09	FR-08	Risk score calculation	Submit incident A (High, 75 records) and incident B (Low, 5 records)	Risk_Score is populated for both; A is not lower than B	Risk score correctly calculated and populated in the breach incident record.	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-10	FR-09	Prediction model test	In the model test screen enter Affected Records 75, Not Contained, Unauthorized Access, High	Predicted Risk Category High	Predicted risk category returned as High with 93% confidence.	Pass
TC-11	FR-09	Prediction field on submission	Submit a breach incident	Predicted_Risk_Category is populated on the saved record	Predicted risk category successfully populated in the submitted breach incident record.	Pass
TC-12	FR-10	High-risk e-mail alert	Submit an incident that the model classifies as High	E-mail with subject "URGENT: High-Risk Breach Incident Detected — ConsentLedger" and incident details is received	High-risk email notification successfully received.	Pass
TC-13	FR-10	No alert for other categories	Submit an incident that the model classifies as Medium or Low	No alert e-mail is sent	No email notification sent for Medium or Low predicted risk categories.	Pass
TC-14	FR-10	Sender address validation	Configure Send Email with a From address that is not an administrator, logged-in or verified user, then with a recognised one	The first configuration is rejected; the second works	Sender-address validation issue resolved by selecting a recognised sender address.	Pass
TC-15	FR-11	Rule configuration	Add or edit a record in Rule_Configuration	Record saved	Rule configuration record successfully saved.	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-16	FR-12	Report display	Open the report of each form after the tests above	Records of each form are listed	Records successfully displayed in their respective reports.	Pass
TC-17	FR-13	Dashboard display	Open the dashboard after the tests above	KPI cards and charts reflect the stored records	Dashboard KPI cards and charts successfully displayed application data.	Pass
TC-18	NFR-02	Unauthorised access	Open the application as a user without access	Access is denied or data is hidden	Unauthorised user access denied or application data hidden.	Pass

Note on TC-10. The model returned a displayed confidence score of 93. This number is the output of the model for one input. It is not a validated probability of harm.

6.3 Debugging Deluge Scripts

Debugging combined the techniques that Zoho Creator provides. Values were inspected by writing them to the info log from a script, scripts were run from the Deluge editor, the workflow configuration was re-checked in the builder, and error messages shown when saving a workflow or submitting a form were read and acted on.

Two real problems were found during development. The first was that the prediction model was not available in the account because of a model reference and configuration issue, so the Predicted_Risk_Category field could not be created through the deployment process. The second was that the Send Email action of the ML_High_Risk_Email_Alert workflow was rejected, because Zoho Creator requires the From address to belong to an administrator, a logged-in user or a verified user. Selecting a sender recognised by Zoho Creator removed the error, and the notification was then tested successfully.

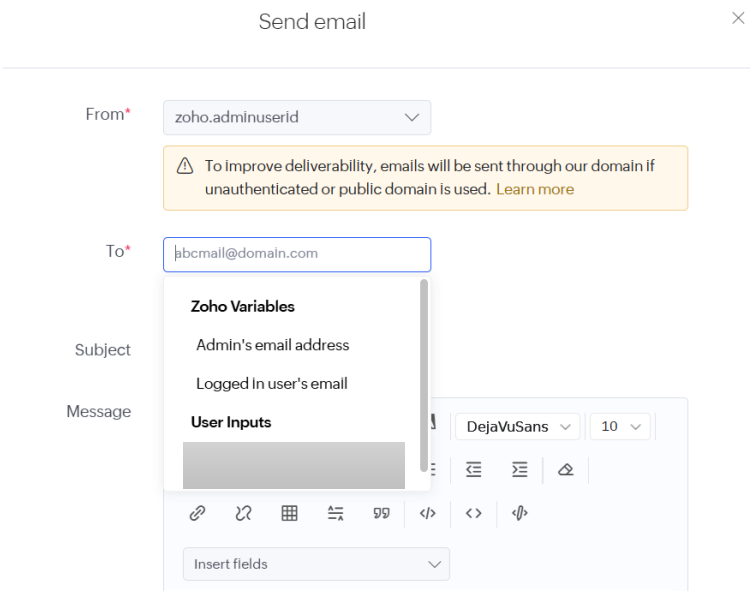


Figure 6.1: Sender-Address Validation Error Message in Zoho Creator

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	The prediction model was not available in the account, so it could not be deployed to Breach_Incidents.	Model reference and configuration issue.	The model reference and configuration were corrected; the model was then published and deployed and Predicted_Risk_Category was populated on submission Not verified in the current application evidence (confirm exact step).	Closed
BUG-02	The Send Email action was rejected with a sender-address validation error.	The From address was not an administrator, a logged-in user or a verified user.	A valid sender recognised by Zoho Creator was configured. The notification was then tested successfully.	Closed

The synthetic nature of the training data is a limitation of the prototype and not a defect. It is discussed in Sections 5.9.2 and 8.4.

6.5 User Acceptance Testing

Informal user acceptance feedback was recorded from two users, as summarised in Table 6.3. The feedback covers navigation, dashboard visibility, incident submission and the risk-assessment workflow. This limited feedback is not a formal or statistically representative UAT study.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Application Administrator	The application interface is simple and easy to navigate. The dashboard provides a clear overview of consent records, rights requests, and breach incidents.	Verified the dashboard and reports. Minor interface improvements were incorporated.
User 2	Assigned Officer	The breach incident form is easy to use. The risk score calculation and predicted risk category help in identifying high-risk incidents.	Tested incident submission and risk prediction. Verified that high-risk incidents trigger email notifications.

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

ConsentLedger has been developed and tested in the development environment of the Zoho Creator account. It has not been published to production, and production deployment is not claimed. The recommended path before any real use is to execute the remaining test cases of Table 6.1, validate the prediction model with representative data, remove test records, configure users and roles, review the privacy and legal requirements with qualified professionals, and then publish with version notes. Rolling back would be done by redeploying the previous version.

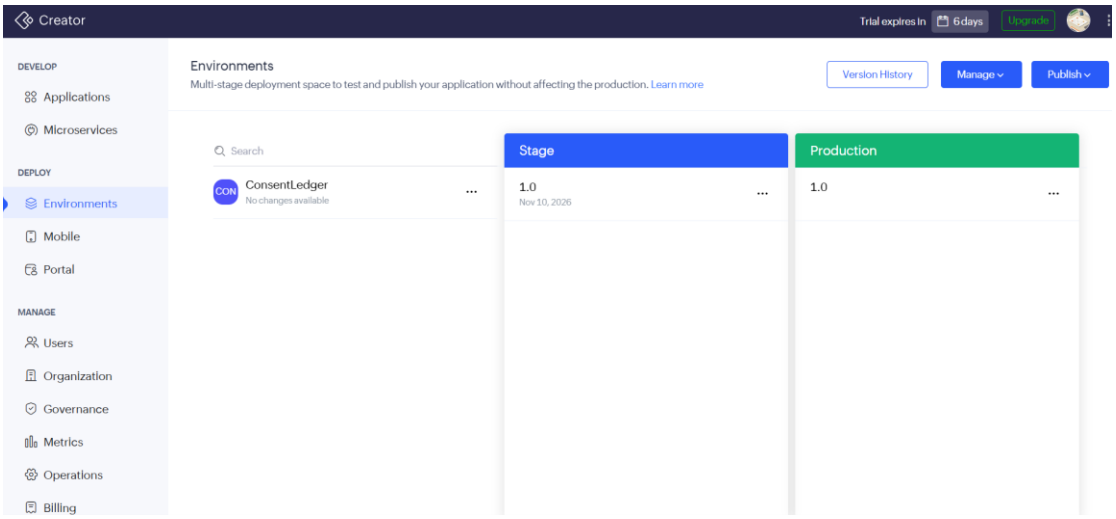


Figure 7.1: Application Environment and Publishing Settings

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	Partly (model test, prediction field population, high-risk e-mail and sender validation)
2	Sample/test data removed or isolated	Not done
3	Roles, permissions and users configured in Production	Not done
4	Connections authorised in Production	Not applicable (no connections)
5	Schedules and workflows enabled	Done in Development (ML_High_Risk_Email_Alert)
6	Application published and URL verified	Not done

S.No.	Checklist Item	Done (✓)
7	Version / release notes recorded	Not done
8	Prediction model validated with representative real-world data	Not done

7.2 Performance Optimisation

No performance measurement was carried out, and none is claimed. The data volume of the prototype is small. The design choices that help performance are that the alert e-mail is sent only for High predictions, so the notification step is skipped for other incidents, and that reports and the dashboard read stored values such as Risk_Score and Predicted_Risk_Category instead of recalculating them. If scripts are extended, record fetches should use criteria and should not be placed inside loops.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Conditional e-mail (High only)	ML_High_Risk_Email_Alert	Not measured	Not measured	Not measured

7.3 Security Features

Security features must be separated into capabilities that the Zoho Creator platform offers and controls that were explicitly implemented in ConsentLedger. Platform capabilities include sign-in, users, roles and permission profiles, and the rule that the sender of an e-mail must be a recognised user. Within ConsentLedger, the Assigned_Officer field restricts the officer to an existing user of the account, and the alert workflow uses a sender that Zoho Creator recognises. Encryption of fields, record-level criteria, two-factor authentication and portal security are not claimed.

Table 7.3: Security Controls

Security Control	Where Applied	Threat Mitigated	Type
Zoho sign-in	Whole application	Anonymous access	Platform capability
User field for officer	Breach_Incidents → Assigned_Officer	Assignment to an unknown person	Implemented in ConsentLedger
Verified sender rule	Send Email action	Mail sent from an unrecognised address	Platform rule; sender configured in ConsentLedger

Security Control	Where Applied	Threat Mitigated	Type
Role and permission profiles	Not established	Unauthorised data access	Not claimed
Record-level criteria	Not configured	Users viewing others' records	Not claimed
Synthetic or fictitious test data	All testing	Exposure of real personal data	Fictitious data used for testing; this is a testing practice, not a technical security control.

Because ConsentLedger is designed to hold personal data, any real use would require a security and privacy review, restricted access, and a data retention decision. These have not been performed.

7.4 Governance and Administration

The application is administered by the application administrator. Form and field names follow a consistent pattern with words joined by underscores, such as Breach_Incidents and Predicted_Risk_Category. Action_Log records selected actions, including consent creation, which helps review but is not an immutable audit trail. Audit-trail settings, change history, backup and export, and user licence management have not been set beyond the platform defaults and are not claimed. The prediction model has its own governance need: it should be re-evaluated and retrained when representative, validated data is available, and its output should be reviewed by a person before any decision is taken.

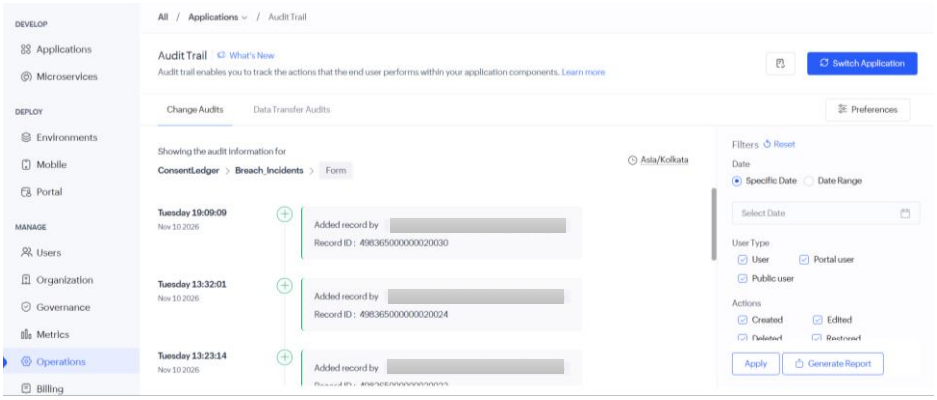


Figure 7.2: Administration and Audit Settings

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The following scenario follows one breach incident through the application. An authorised user discovers unauthorised access to a system and opens the Breach_Incidents form. The user enters the title, description, discovery date and time, severity High, an estimated 500 affected records, an assigned officer and a response due date, and submits the form (Figure 8.1). The Risk_Score is calculated, and the prediction model populates Predicted_Risk_Category. The submitted incident is shown in Figure 8.2. The separate model test described in Section 5.9.2 used 75 affected records and returned High with a displayed confidence score of 72.20. The high-risk e-mail notification was demonstrated using the test incident shown in Figure 8.3; its incident reference and severity differ from the submitted incident in Figures 8.1 and 8.2. The assigned officer then investigates, and the incident status and resolution summary are updated by a person.

The model test returned High with a displayed confidence score of 72.20, and the high-risk e-mail notification was tested successfully using a separate test incident. These are the confirmed results. Counts, processing times and predictive performance on real-world data were not measured.

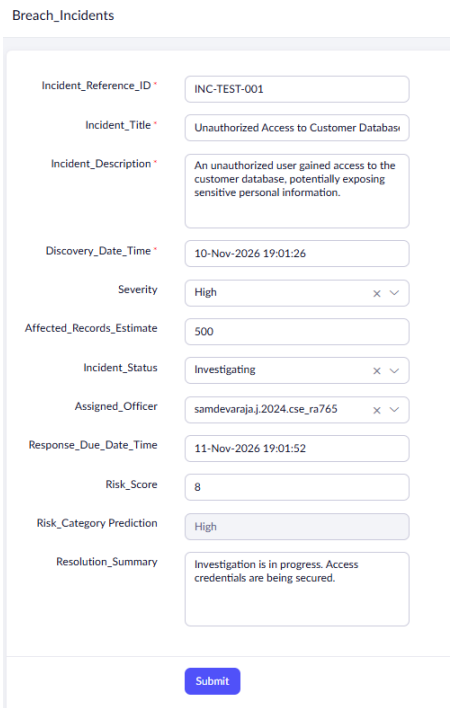


Figure 8.1: Scenario Step 1 – User Submits a Breach Incident

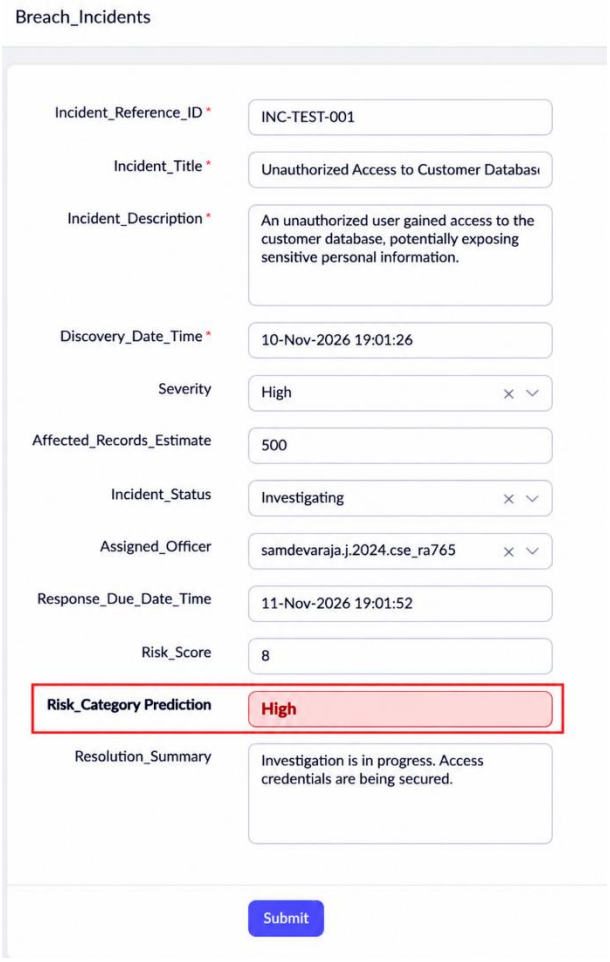


Figure 8.2: Scenario Step 2 – Predicted_Risk_Category Shows High

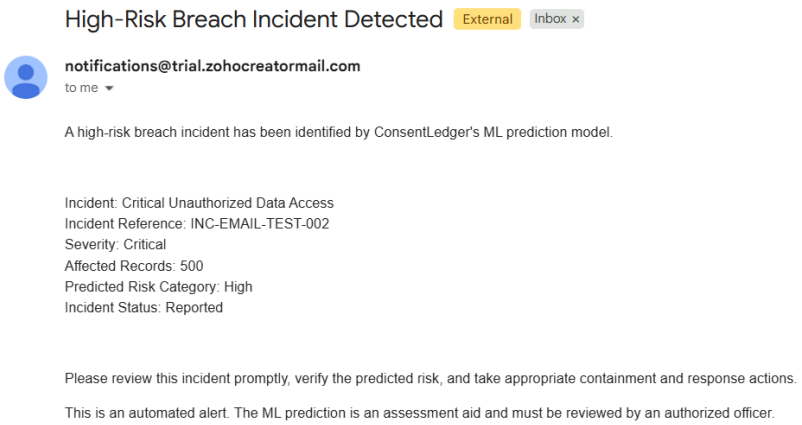


Figure 8.3: Scenario Step 3 – High-Risk Alert E-mail Received

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Data model and forms	Sections 4.2, 4.3, 5.2	Six forms implemented; field specifications are documented in Tables 4.3–4.8.
Automation using Deluge	Section 5.5, Appendix C	Deluge used for classification, deadline risk, breach risk score and consent-creation logging; source code is included in Appendix C.
Prediction model	Section 5.9	Model trained on 100 synthetic records and deployed; test returned High (72.20); field populated on submission.
Conditional e-mail alert	Section 5.6	ML_High_Risk_Email_Alert configured and tested successfully for High predictions.
Reports and dashboard	Sections 5.3, 5.4	Reports and a dashboard with KPI cards and charts built.
Testing and honest limitations	Chapters 6, 7, 8	Four tests are recorded as executed; remaining test cases are identified as not executed. Synthetic-data limitations are stated.

8.3 Comparison with the Existing System

Time per request, productivity gains and similar measures were not taken, so no improvement figures are claimed. Table 8.2 compares the two systems qualitatively.

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Not measured	Not measured	Not measured
Record keeping	Spreadsheets, e-mail and paper	Six forms in one application	One place for records (qualitative)
Incident assessment	Subjective judgement	Rule-based Risk_Score plus model prediction	Repeatable, with a prediction that needs validation
Escalation	Depends on a person remembering	E-mail alert for High predictions	Less dependence on memory; does not replace human response

Parameter	Existing System	Proposed Application	Improvement
Monitoring	Manual counting	Reports and dashboard	Summary on demand (qualitative)

8.4 Limitations of the Low-Code Approach

The limitations seen in this project are of two kinds. The first kind concerns the platform. The appearance of forms and dashboards is limited to the options of the builder, complex logic spread over several forms is easier to maintain in Deluge than in visual configuration, and the application depends on the vendor and on the plan used [7]. The built-in prediction model also offers limited control over the algorithm, and the model had to be available in the account before it could be deployed, which caused the first defect of Section 6.4.

The second kind concerns the machine-learning component and is more important. The training data contained 100 synthetic records whose labels were generated using illustrative rules. A model trained on such data learns the rules that produced the labels and not the behaviour of real incidents. Its displayed 100% training accuracy shows that it reproduced the rules, which is expected, and says nothing about real-world accuracy. The displayed confidence score is not a validated probability of harm. The dataset is small, and the class balance and the coverage of unusual incidents are not established. For operational use the model would have to be retrained on representative, validated breach data and independently evaluated on data it has not seen, and its output would have to remain advisory.

The application also does not claim legal compliance. Whether any process satisfies the DPDP Act [1] depends on legal interpretation, the applicable rules and the practices of the organisation, and not on the software alone.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of ConsentLedger is no-code configuration: forms, fields, reports, the dashboard, the user field for the assigned officer, the deployment of the prediction model and the e-mail workflow. Low-code Deluge scripting was needed for the logic that configuration alone could not express, namely request classification, deadline-risk assessment, risk score calculation and action logging. Pro-code was not used. It would be considered if the organisation needed advanced backend processing, model

retraining pipelines, integration with an incident management system or a custom mobile application. For a prototype of this size the no-code and low-code parts were sufficient.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project built ConsentLedger, a consent lifecycle and data principal rights management prototype, on Zoho Creator. It has six forms for persons, consent, rights requests, breach incidents, action logging and rule configuration, with reports and a dashboard. Deluge scripts classify rights requests, assess deadline risk, calculate a breach risk score and log selected actions. A prediction model trained on 100 synthetic records was deployed on the Breach_Incidents form, and a workflow sends an e-mail alert when the predicted risk category is High. The model test and the e-mail notification were completed successfully, and two defects met during development were resolved.

The work is a prototype. It has been tested in the development environment only, its training data is synthetic, and no claim of legal compliance, production readiness or immutable audit logging is made. Within those limits it shows that a low-code platform can combine record keeping, rule-based scoring, a machine-learning prediction and conditional notification in one application with little hand-written code.

9.2 Future Enhancements

The following improvements are proposed. They are enhancements and not completed functionality.

1. Integrate Zoho Catalyst for advanced backend processing.
2. Use Zoho Analytics for advanced privacy operations dashboards.
3. Add automated deadline reminders and escalation for rights requests.
4. Implement more comprehensive role-based access control with separate profiles and record-level criteria.
5. Integrate with external incident management systems.
6. Retrain the prediction model on representative, validated datasets and evaluate it independently.
7. Enhance audit logging and monitoring, including protection against edits to log records.

8. Automate more of the privacy request lifecycle.
9. Deploy to production with appropriate security and governance controls after a legal and security review.

9.3 Learning Outcomes and Career Relevance

Through this project the student learned to design a data model and forms, write Deluge logic for classification, scoring and logging, train and deploy a prediction model inside a low-code platform, configure a conditional e-mail workflow, and diagnose platform errors such as the sender-address validation rule. The project also taught the importance of stating the limits of a model honestly, in particular when the training data is synthetic. These skills are relevant to roles such as citizen developer, low-code developer, solutions consultant and automation engineer, and to privacy operations work where data handling and escalation processes have to be supported by tools.

REFERENCES

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[3] W. Rosam and O. Kalnichevski, *Low-Code Development: A Practical Guide*. New Delhi, India: BPB Publications.

[4] Zoho Corporation, "Zoho Creator Help Documentation." (Online). Available: https://www.zoho.com/creator/help/ (Accessed: November 2026).

[5] Zoho Corporation, "Deluge Script Reference Guide." (Online). Available: https://www.zoho.com/deluge/ (Accessed: November 2026).

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[7] Y. Luo, P. Liang, C. Wang, M. Shahin, and J. Zhan, "Characteristics and challenges of low-code development: The practitioners' perspective," in *Proc. 15th ACM/IEEE Int. Symp. Empirical Software Engineering and Measurement (ESEM)*, 2021.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

The checklist below records the feature coverage evidenced in the current prototype. Each feature is marked Implemented, Partially implemented or Not implemented. Features for which no implementation evidence is available are conservatively classified as Not implemented or not evidenced; the live application should be demonstrated against this checklist.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Implemented	ConsentLedger application and navigation menu	5.1	5.1
II	Forms with ≥ 8 distinct field types	Not implemented	Not implemented or not evidenced in the current prototype.	5.2	–
II	Lookup field (relationship between forms)	Implemented	Person lookup relationships in Consent_Records and Rights_Requests.	4.2, 5.2	–
II	Subform (parent–child data)	Not implemented	Not implemented or not evidenced in the current prototype.	4.2, 5.2	–
II	Field validations / layout rules	Not implemented	Not implemented or not evidenced in the current prototype.	5.2	–
II	List report with filter, grouping, custom action	Partially implemented	Reports on the forms; filters, grouping and actions to confirm	5.3	5.8
II	Kanban / Calendar / Timeline / Map report	Not implemented	Not implemented or not evidenced in the current prototype.	5.3	–
II	Pivot table / chart report	Partially implemented	Charts on the dashboard; separate chart reports to confirm	5.3, 5.4	5.9
II	Page / dashboard built with UI Builder	Implemented	Dashboard with KPI cards and charts	5.4	5.9

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Deluge – form events (validate / success / field actions)	Implemented	Scripts DS-01 to DS-04; form event triggers described in Section 5.6.	5.5	5.10
II	Deluge – functions, fetch, loops, collections	Not implemented	Not implemented or not evidenced in the current prototype.	5.5	–
II	Email notification	Implemented	Conditional e-mail through ML_High_Risk_Email_ Alert; no in-app notification is claimed.	5.5, 5.6	5.14
II	Form workflow	Implemented	ML_High_Risk_Email_ Alert	5.6	5.13
II	Scheduled workflow	Not implemented	No scheduled reminder or escalation	5.6	–
II	Approval process	Not implemented	Not part of the prototype	5.6	–
II	Blueprint	Not implemented	Not used	5.6	–
II	Batch / report workflow	Not implemented	Not used	5.6	–
II	Testing & debugging evidence	Partially implemented	Model test, e-mail test and two defects resolved; other cases pending	6	6.1
III	Integration with a Zoho service	Not implemented	Catalyst and Analytics not integrated	5.7	–
III	Connection / REST API call (invokeurl)	Not implemented	Not used	5.7	–
III	Webhook or Creator REST API	Not implemented	Not used	5.7	–
III	Users, roles and permission profiles	Partially implemented	User field for officer; role profiles not established	5.8	5.15

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Record-level access criteria / customer portal	Not implemented	No record-level criteria configured; portal URL exists but public portal access is not verified.	5.8	–
III	Zia / AI feature	Implemented	Zoho Creator prediction model ConsentLedger_Breach_Risk	5.9	5.16
III	Analytics (charts / Zoho Analytics)	Partially implemented	Dashboard charts and KPI cards; Zoho Analytics not connected	5.9	5.9
IV	Performance optimisation applied & measured	Not implemented	Not measured	7.2	–
IV	Deployment across environments	Not implemented	Development only	7.1	–
IV	Security controls (encryption, OAuth, permissions)	Partially implemented	Sign-in, user field and sender rule only	7.3	–
IV	Governance (audit trail, administration)	Partially implemented	Selected action logging; not an immutable audit trail	7.4	–
V	Pro-code extension (Catalyst / widgets / custom functions)	Not implemented	Not used	5.10	–
V	Low-code limitations & spectrum analysis	Implemented	Written discussion	8.4, 8.5	–
—	Mobile app / portal access	Not implemented	Not customised	5.11	–

Total features implemented: 13 / 32 fully, with 7 partially implemented.

APPENDIX B – COURSE OUTCOME MAPPING

The table maps each course outcome of CB23F35 to the chapters of the report. The self-assessed level uses the scale 1 – Low, 2 – Medium, 3 – High, and the examiner may revise it. The outcome statements are those given in the faculty template.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3 – High	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	2 – Medium	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	1 – Low	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	1 – Low	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2 – Medium	

CO2 is self-assessed as Medium because forms, reports, Deluge scripts and a workflow were built, but approval, schedule and Blueprint workflows were not. CO3 and CO4 are self-assessed as Low because no external integration, full role-based access or production deployment was implemented. CO5 is Medium because the AI prediction model and the limitations of low-code are evidenced, while pro-code extension was not used. CO1 is High because the principles, architecture and justification are documented in Chapters 1, 2 and 4.

APPENDIX C – COMPLETE DELUGE SOURCE CODE

This appendix reproduces the verified Deluge source code for DS-01 (rights request classification), DS-02 (deadline-risk assessment), DS-03 (breach risk-score calculation) and DS-04 (consent-creation logging). DS-05 is omitted because no separate working script for other selected-action logging has been verified. The ML_High_Risk_Email_Alert is a workflow configuration and is documented separately; it is not a Deluge source-code listing.

C.1 DS-01 – Rights Request Classification

```
// DS-01: Rights request classification
requestText = ifnull(input.Request_Text,"").toLowerCase();
suggestedType = "Other";
if(requestText.contains("access")      ||      requestText.contains("copy")      ||
requestText.contains("provide    my    data")    ||    requestText.contains("personal
information"))
{
    suggestedType = "Access";
}
else   if(requestText.contains("correct")   ||   requestText.contains("update")   ||
requestText.contains("inaccurate"))
{
    suggestedType = "Correction";
}
else   if(requestText.contains("erase")   ||   requestText.contains("delete")   ||
requestText.contains("remove my data"))
{
    suggestedType = "Erasure";
}
input.AI_Suggested_Type = suggestedType;
input.Classification_Status = "Pending Review";
```

C.2 DS-02 – Deadline-Risk Assessment

```
// DS-02: Rights request deadline-risk assessment
if(input.Request_Status != "Completed" && input.Request_Status != "Closed" &&
input.Request_Status != "Rejected")
{
    if(input.Due_Date == null)
    {
        input.Delay_Risk_Level = "Not Assessed";
        input.Risk_Reason = "Due date is missing; review required.";
    }
    else
    {
```

```
        daysLeft = daysBetween(zoho.currentdate,input.Due_Date);
        if(daysLeft < 0)
        {
            input.Delay_Risk_Level = "High";
            input.Risk_Reason = "Request is overdue.";
        }
        else if(daysLeft <= 2)
        {
            input.Delay_Risk_Level = "Medium";
            input.Risk_Reason = "Due date is within 2 days.";
        }
        else
        {
            input.Delay_Risk_Level = "Low";
            input.Risk_Reason = "More than 2 days remain before the due date.";
        }
    }
}
else
{
    input.Delay_Risk_Level = "Not Assessed";
    input.Risk_Reason = "Request is closed or completed; no pending deadline
risk.";
}
```

C.3 DS-03 – Breach Risk Score Calculation

```
// Script ID : DS-03
riskScore = 0;
if(input.Severity == "Low")
{
        riskScore = 1;
}
else if(input.Severity == "Medium")
{
        riskScore = 3;
}
else if(input.Severity == "High")
{
        riskScore = 6;
}
else if(input.Severity == "Critical")
{
        riskScore = 8;
}
if(input.Affected_Records_Estimate != null && input.Affected_Records_Estimate >=
100)
{
        riskScore = riskScore + 2;
```

```
}
else if(input.Affected_Records_Estimate != null && input.Affected_Records_Estimate
>= 10)
{
        riskScore = riskScore + 1;
}
if(riskScore > 10)
{
        riskScore = 10;
}
input.Risk_Score = riskScore;
```

C.4 DS-04 – Consent Creation Logging

```
// Script ID : DS-04
uniqueID = "Consent_Created_" + input.ID + "_" +
zoho.currenttime.toString("yyyyMMddHHmmss");
insert into Action_Log
[
        Action_Reference_ID=uniqueID
        Action_Date_Time=zoho.currenttime
        Action_Type="Consent Created"
        Related_Record_Type="Consent"
        Related_Record_ID=input.ID
        Action_Details="A new consent record was created with ID " + input.ID
        Added_User=zoho.loginuser
]
```

C.5 ML_High_Risk_Email_Alert (workflow configuration)

```
// Workflow     : ML_High_Risk_Email_Alert
// Form         : Breach_Incidents
// Record event : Created
// Form event   : Successful form submission (On Success)
// Condition    : Predicted_Risk_Category equals High
// Action       : Send Email
//   From    : sender recognised by Zoho Creator
//             (administrator, logged-in or verified user)
//   To      : Configured recipient: project owner's authorised
institutional email address (redact in public copies)
//   Subject : URGENT: High-Risk Breach Incident
//             Detected — ConsentLedger
//   Message : inserts Incident_Title,
//             Incident_Reference_ID, Severity,
//             Affected_Records_Estimate,
//             Predicted_Risk_Category,
//             Incident_Status
```

APPENDIX D – USER MANUAL

These instructions describe the features that are available in ConsentLedger. Screen names must be matched with the menu of the live application. No credentials or access URL are printed in this manual.

D.1 Accessing the Application

1. Open a web browser and go to the application URL Open ConsentLedger from the authorised Zoho Creator account.
2. Sign in with your Zoho account when asked. Your administrator must have given you access.
3. The menu shows the forms, reports and the dashboard of ConsentLedger.

D.2 Instructions for the Application Administrator

1. To add users, invite them to the application from the Zoho Creator user settings so that they can be selected as Assigned Officers.
2. To review rules, open Rule_Configuration and view or edit the records that hold the operational rules.
3. To monitor the application, open the dashboard and read the KPI cards and charts.
4. To review actions, open the Action_Log report and check the entries.
5. To review incidents, open the Breach_Incidents report and read Risk_Score and Predicted_Risk_Category together. Treat the prediction as advice and confirm it by investigation.

D.3 Instructions for Authorised Application Users

1. To register a person, open Person_Register, enter the details and submit.
2. To record consent, open Consent_Records, select the registered person, enter the consent details and submit. A log entry is added to Action_Log.
3. To record a rights request, open Rights_Requests, enter the request details and submit. The request is classified and its deadline risk is assessed by the application.

4. To record a breach incident, open Breach_Incidents and fill in the title, description, discovery date and time, severity, status, estimated affected records, assigned officer and response due date. Submit the form.
5. After submission, check the Risk_Score and Predicted_Risk_Category on the record. If the prediction is High, an alert e-mail is sent. Alerts are not sent for Medium or Low predictions, so do not wait for an e-mail for those incidents.
6. To view records, open the report of the relevant form from the menu.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SHANTHOSH S.	Register No.	240701487
Project Title	CONSENTLEDGER: A CONSENT LIFECYCLE AND DATA PRINCIPAL RIGHTS MANAGEMENT SYSTEM	Date	11 November 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name: _____	**Signature of External Examiner** Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SHANTHOSH S	Register No.	240701487
Department / Section	Computer Science and Engineering	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____